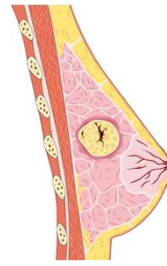
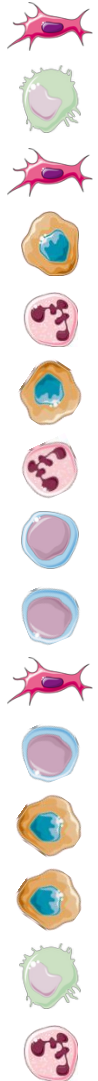


Spatially-Aware Clustering

Ahmed Mahfouz

Human Genetics, Leiden University Medical Center
Pattern Recognition and Bioinformatics, TU Delft

scRNA-seq

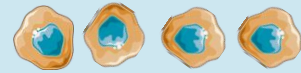


Spatial transcriptomics

scRNA-seq



Spatial transcriptomics



Malignant



Lymphocyte



Fibroblast



Monocyte

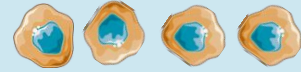


Neutrophil

scRNA-seq



Spatial transcriptomics



Malignant



Lymphocyte



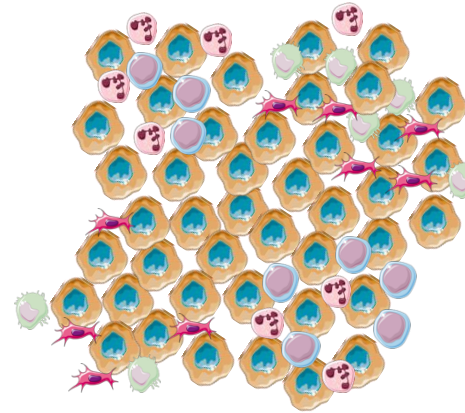
Fibroblast



Monocyte



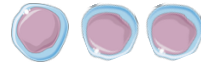
Neutrophil



scRNA-seq



Malignant



Lymphocyte



Fibroblast

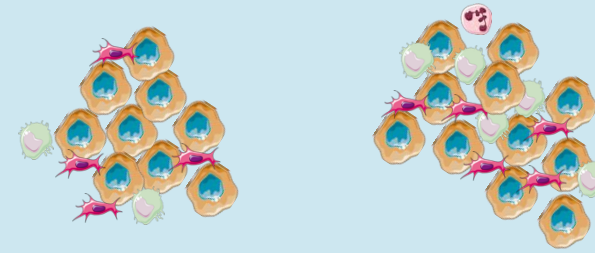


Monocyte



Neutrophil

Spatial transcriptomics



Domain 1



Domain 2



Domain 3

What are niches/domains/... in spatial omics?

Method | [Open access](#) | Published: 12 January 2024

Niche-DE: niche-differential gene expression analysis in spatial transcriptomics data identifies context-dependent cell-cell interactions

» In our analyses, we define a cell's niche as the cell-type composition of its local neighborhood, where the range of the local neighborhood is determined by a kernel function.

12k Accesses | 43 Altmetric | [Metrics](#)

Article | [Open access](#) | Published: 02 April 2024

The covariance environment defines cellular niches for spatial inference

[Doron Haviv](#), [Ján Remšík](#), [Mohamed Gatie](#), [Catherine Snopkowski](#), [Meril Takizawa](#), [Nathan Pereira](#), [John Bashkin](#), [Stevan Jovanovich](#), [Tal Nawy](#), [Ronan Chalique](#), [Adrienne Boire](#), [Anna-Katerina Hadjantonakis](#) &

» The local neighborhood, or niche, of a cell is a useful resolution for defining cell interactions; it may represent functional anatomical subunits (such as stem cell niches) and is a basis for identifying larger spatial patterns.

Brief Communication | [Open access](#) | Published: 27 October 2022

Modeling intercellular communication in tissues using spatial graphs of cells

[David S. Fischer](#), [Anna C. Schaar](#) & [Fabian J. Theis](#) 

[Nature Biotechnology](#) 41, 332–336 (2023) | [Cite this article](#)

» Spatial graphs of cells are based on segmentation masks of cells in spatial molecular profiling data. Resolution is the radius of neighborhood used to define a niche.

Quantitative characterization of cell niches in spatial atlases

 Sebastian Birk,  Irene Bonafonte-Pardàs, [Adib Miraki Feriz](#), [Adam Boxall](#),  Eneritz Agirre, [Fani Memi](#),  Anna Maguza,  Rong Fan,  Gonçalo Castelo-Branco,  Fabian J. Theis,  Omer Ali Bayraktar,

» Identifying niches and quantifying their functions can help to unravel important insights into tissue architecture and spatial biomarkers, which can be harnessed to create novel diagnostic tools, identify new drug targets, and develop targeted therapies.

Nicheformer: a foundation model for single-cell and spatial omics

 Anna C. Schaar,  Alejandro Tejada-Lapuerta,  Giovanni Pallarès,  Mariia Minaeva,  Larsen Vornholz,  Leander Dony, [Francesco](#),  Fabian J. Theis

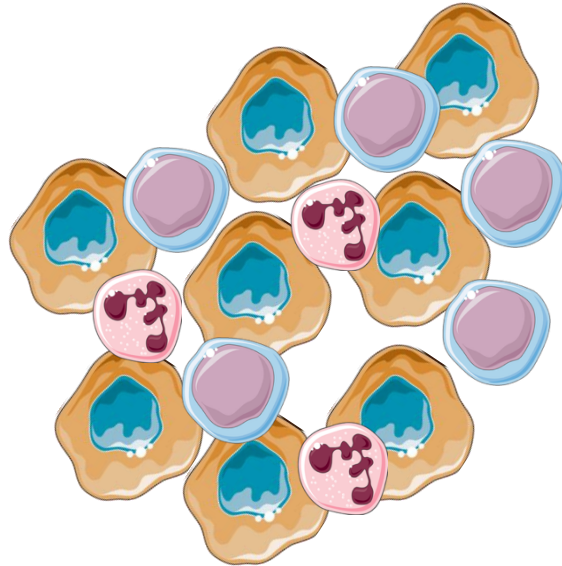
doi: <https://doi.org/10.1101/2024.04.15.589472>

» In contrast to standard approaches to identifying cell types, cellular niches are defined by spatially-dependent labels obtained by combining similarity in gene expression profiles together with spatial neighborhood structures, histological information or global structures in the tissue Cellular niches typically describe a local structure in a tissue, for example, a tumor niche or immune niche, which vary both in terms of gene expression, but also in terms of the cell type compositions. Similarly to niches, tissue regions can be defined as spatially consistent cellular states across a larger-than-a-niche tissue context. Importantly tissue regions are commonly composed of multiple different niches and therefore describe higher-level structures in a tissue.

Let's call them spatially-aware
clusters (SACs)

Spatial clustering principles

Cluster cells based on the gene expression of:



Spatial clustering principles

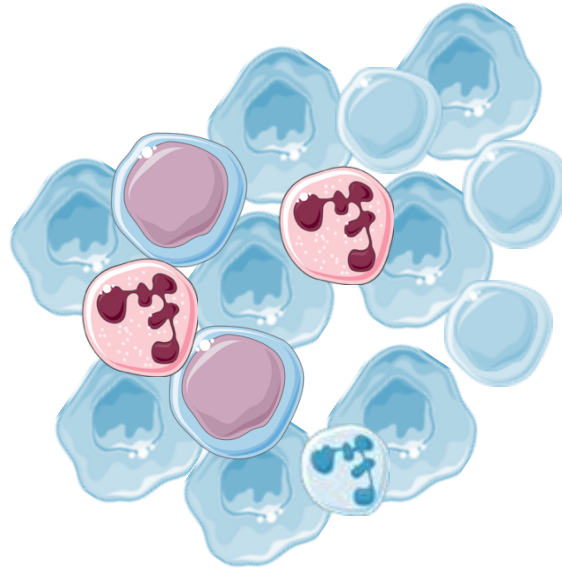
Cluster cells based on the gene expression of:



Cell

Spatial clustering principles

Cluster cells based on the gene expression of:



**Cell
+
Immediate
neighbors
(niche)**

Spatial clustering principles

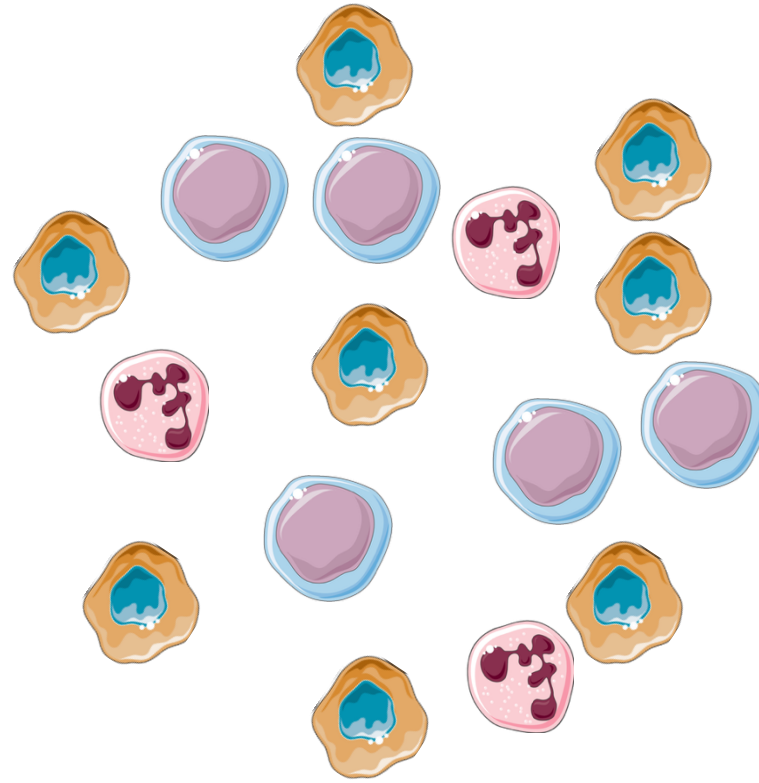
Cluster cells based on the gene expression of:



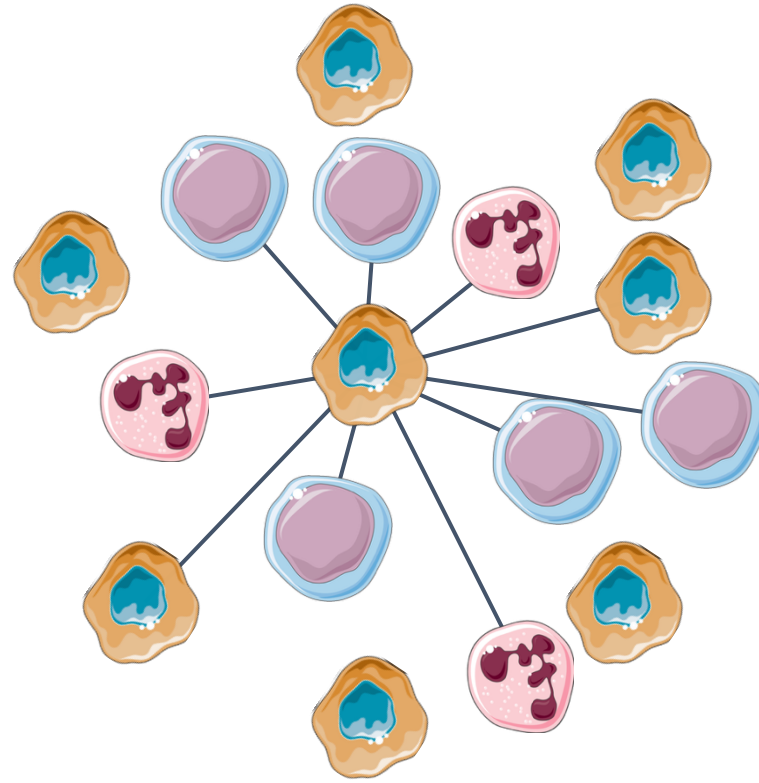
Cell
+
Immediate
neighbors
(niche)
+
Farther neighbors

Spatial domain identification approaches

Approach 1: proportion of cell types



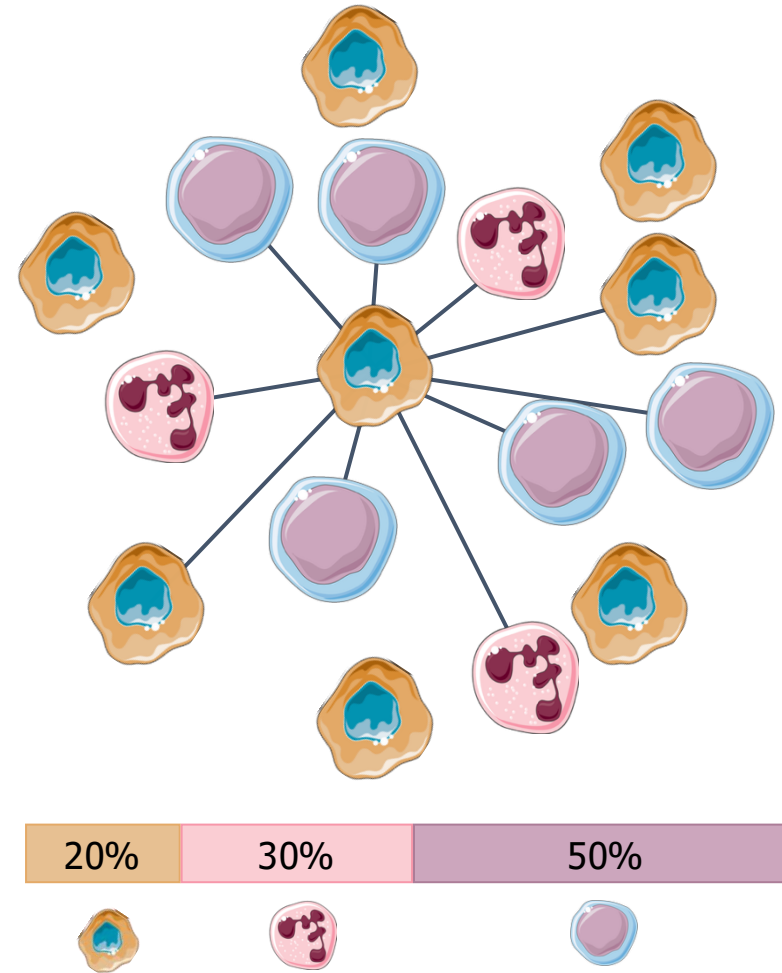
Approach 1: proportion of cell types



Goltsev et al. "Deep profiling of mouse splenic architecture with CODEX multiplexed imaging." *Cell* (2018).

Schürch et al. "Coordinated cellular neighborhoods orchestrate antitumoral immunity at the colorectal cancer invasive front." *Cell* (2020).

Approach 1: proportion of cell types



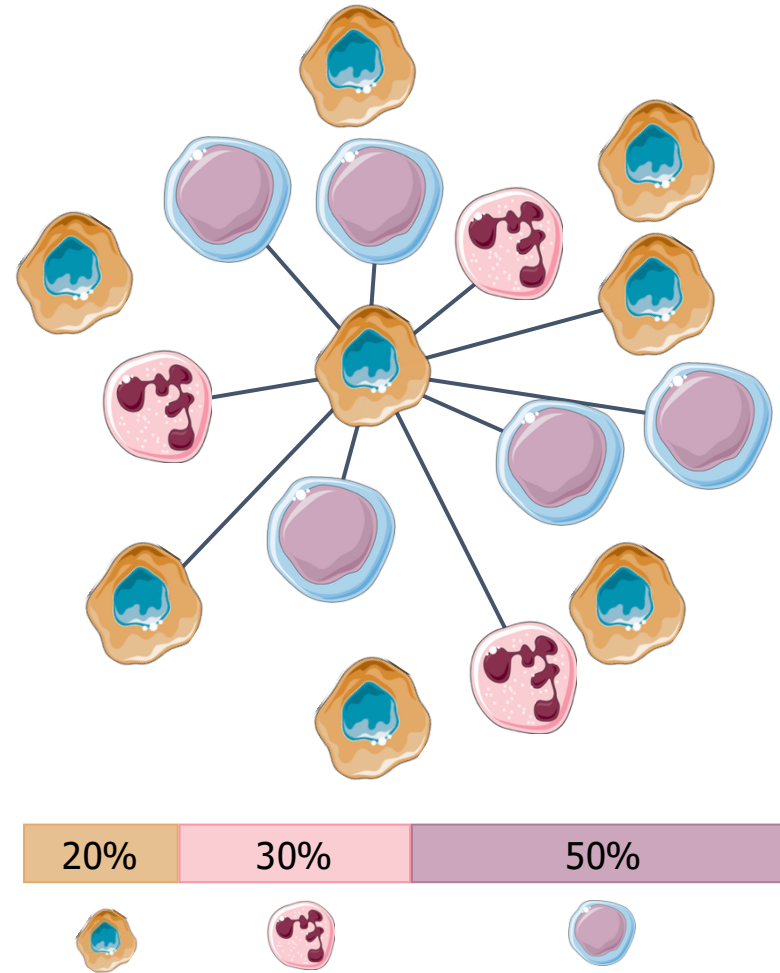
Goltsev et al. "Deep profiling of mouse splenic architecture with CODEX multiplexed imaging." *Cell* (2018).

Schürch et al. "Coordinated cellular neighborhoods orchestrate antitumoral immunity at the colorectal cancer invasive front." *Cell* (2020).

Approach 1: proportion of cell types

Steps

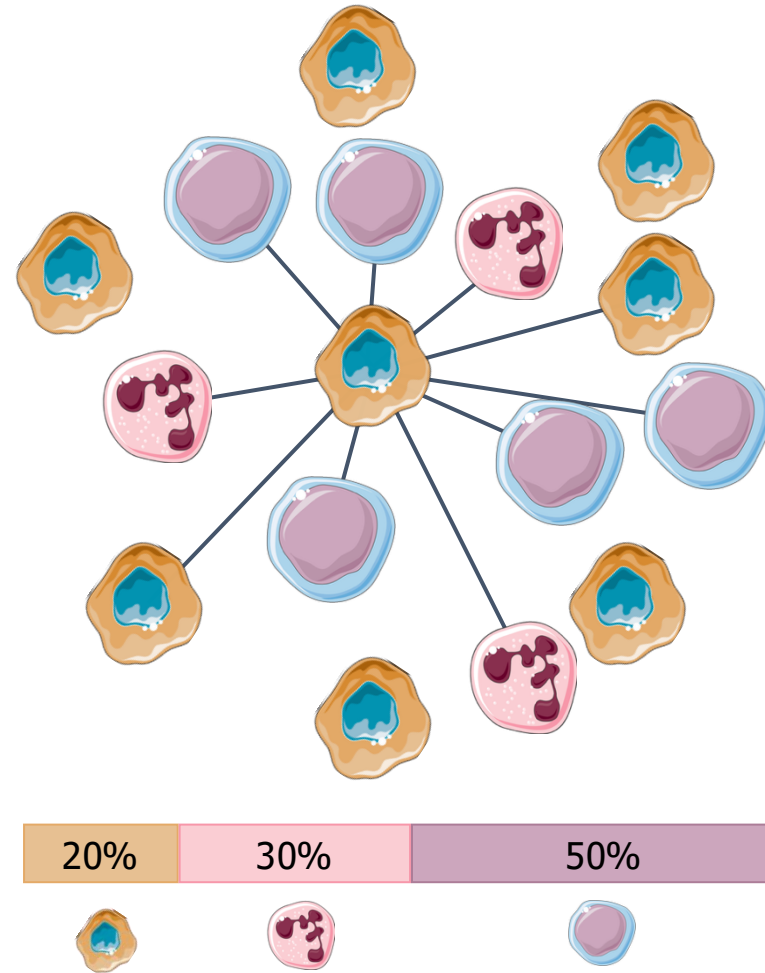
1. Compute neighbors proportion
for every cell
2. Cluster cells
based on their proportions



Approach 1: proportion of cell types

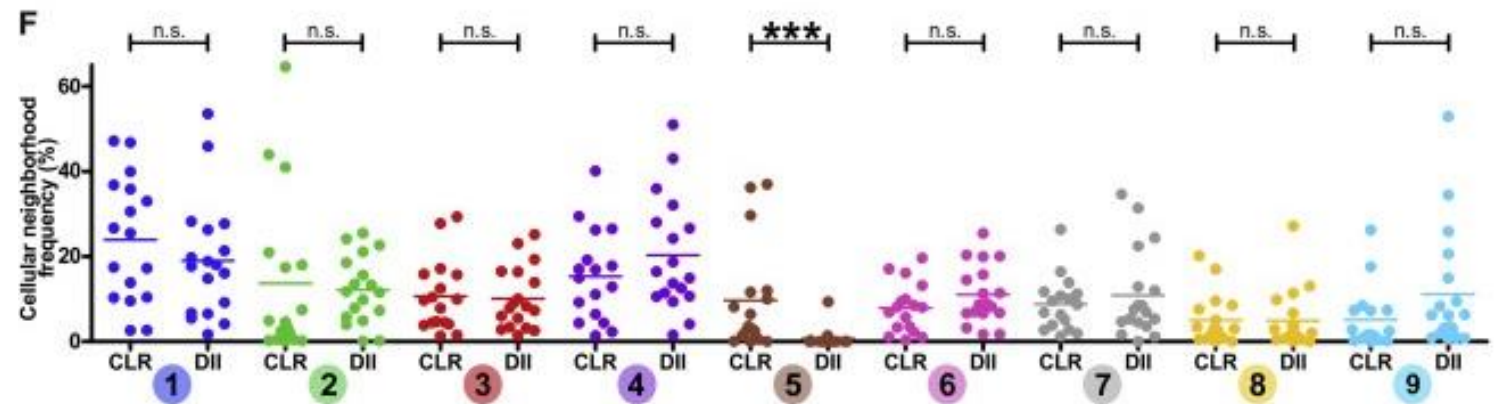
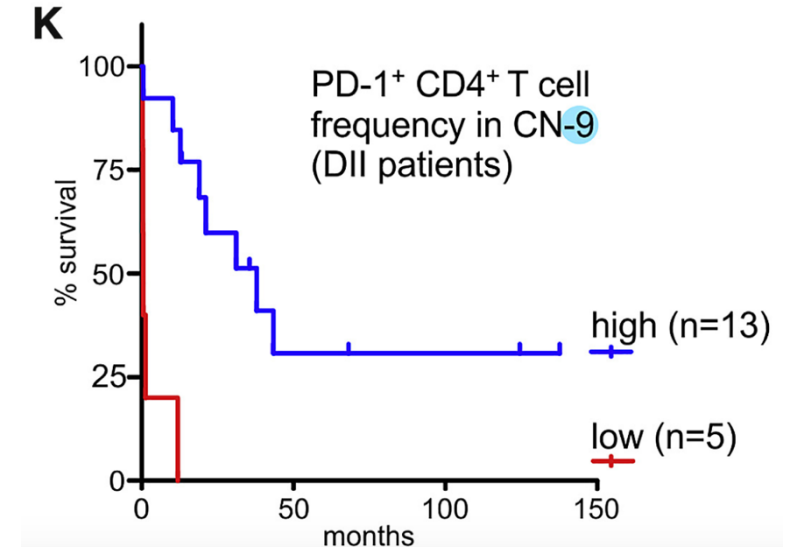
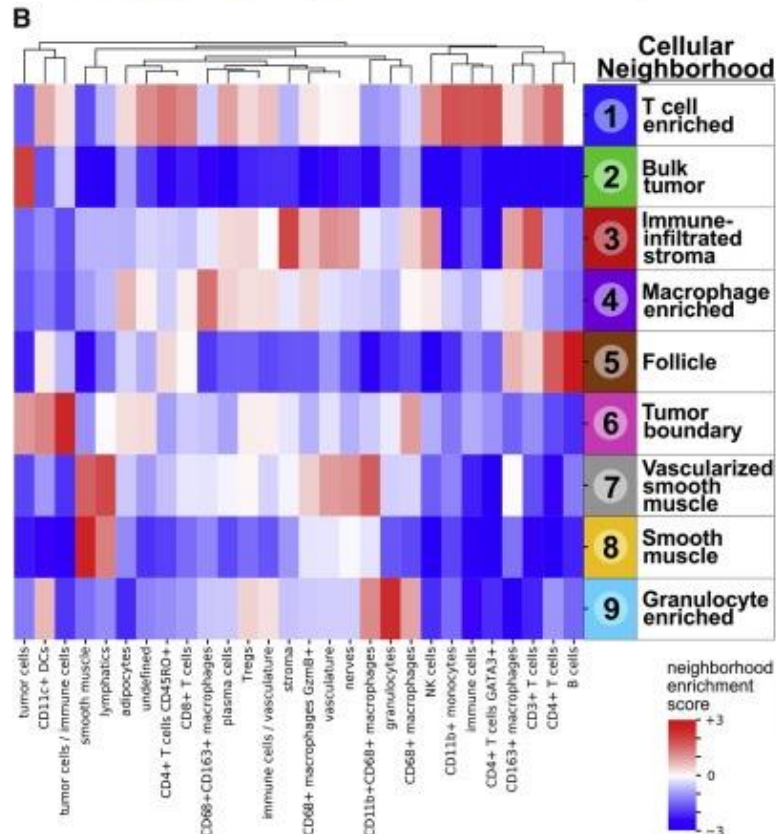
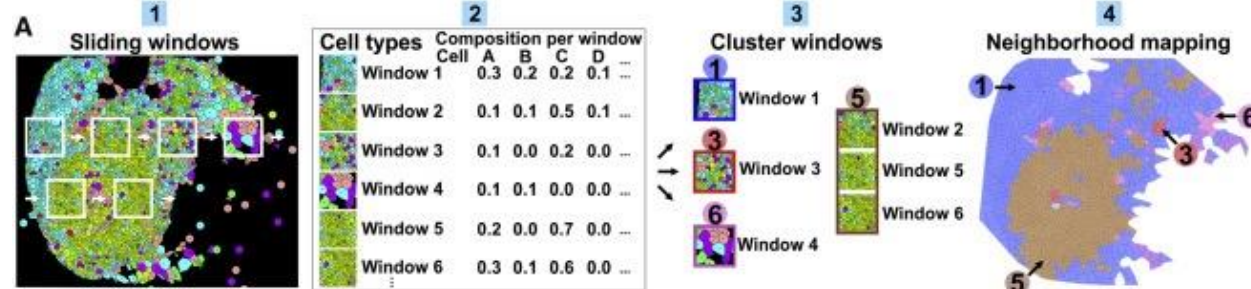
Steps

1. Compute neighbors proportion
for every cell
2. Cluster cells
based on their proportions



- Scalable
- Depends on manual annotations
 - How detailed should the annotation be?
 - How to capture **variability** within cell types?

Approach 1: proportion of cell types

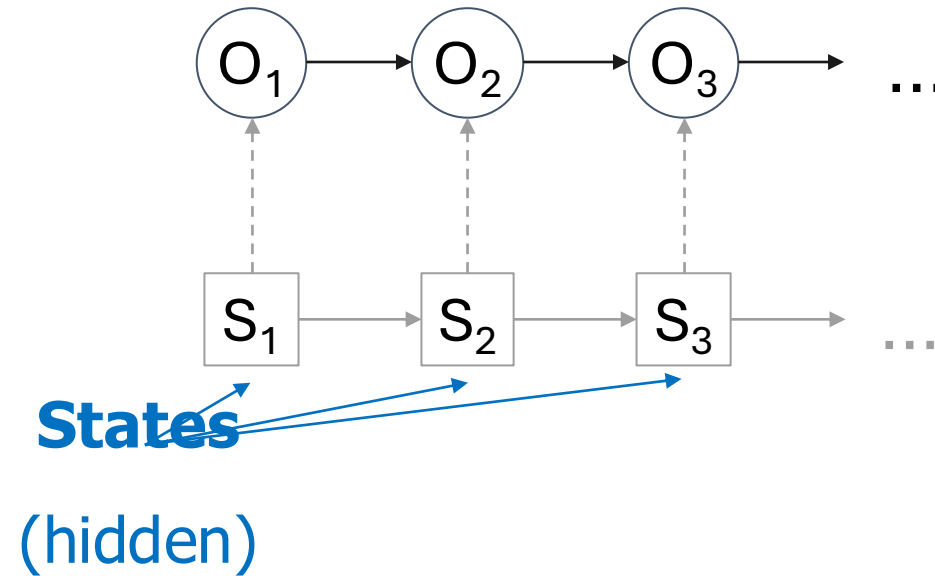


Goltsev et al. "Deep profiling of mouse splenic architecture with CODEX multiplexed imaging." *Cell* (2018).

Schürch et al. "Coordinated cellular neighborhoods orchestrate antitumoral immunity at the colorectal cancer invasive front." *Cell* (2020).

Approach 2: Hidden Markov Random Fields

Hidden Markov Model

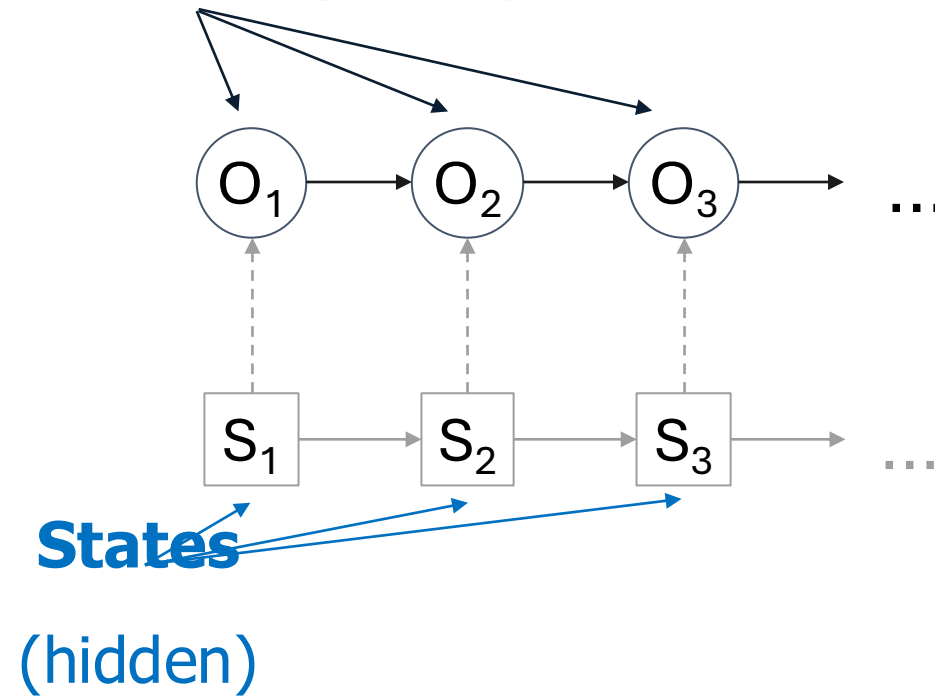


Sequences $\rightarrow$ 1D graphs

Approach 2: Hidden Markov Random Fields

Hidden Markov Model

Observations (visible)



Sequences $\rightarrow$ 1D graphs

Approach 2: Hidden Markov Random Fields

Histone marks

Hidden Markov Model

levels

Observations (visible)

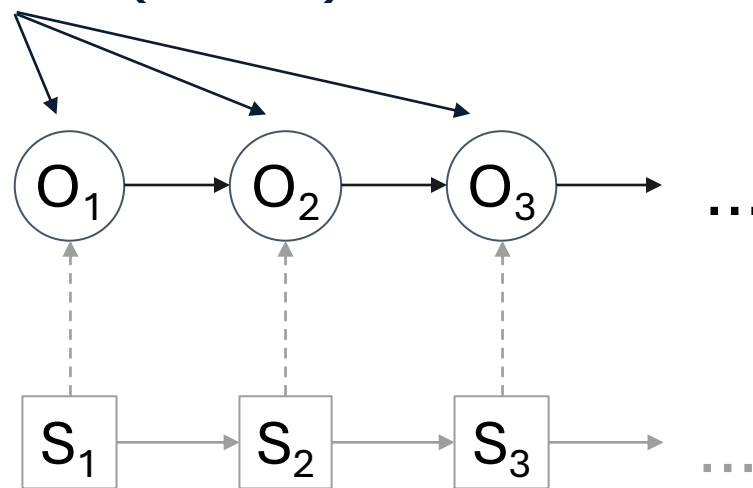
(high vs low)

Chromatin state

(active vs inactive)

States

(hidden)



Genome sequence

Sequences $\rightarrow$ 1D graphs

Approach 2: Hidden Markov Random Fields

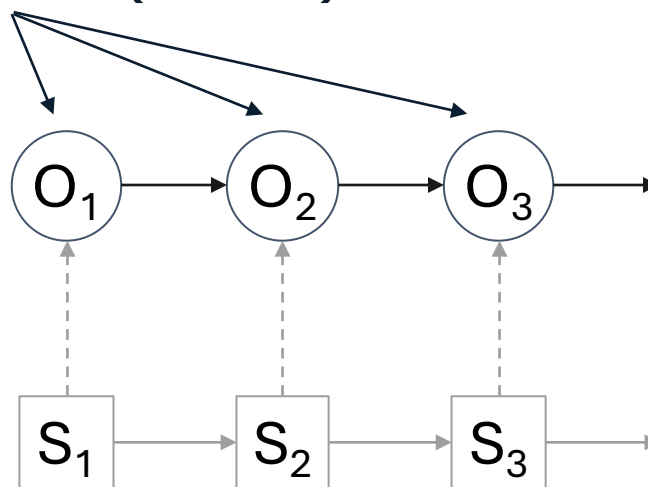
Histone marks

Hidden Markov Model

levels

Observations (visible)

(high vs low)



... **Neighboring observations (O)**

are likely to have

similar states (S)

Chromatin state

(active vs inactive)

States

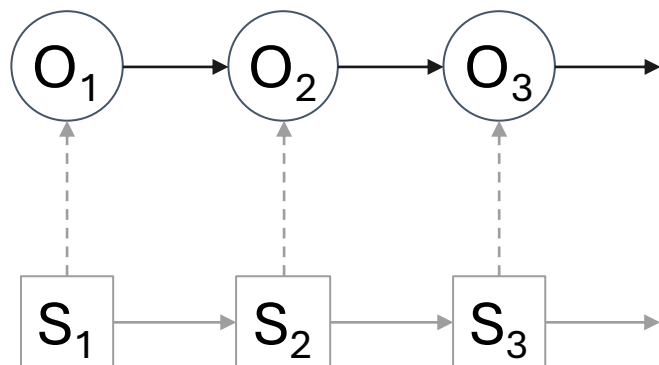
(hidden)

Genome sequence

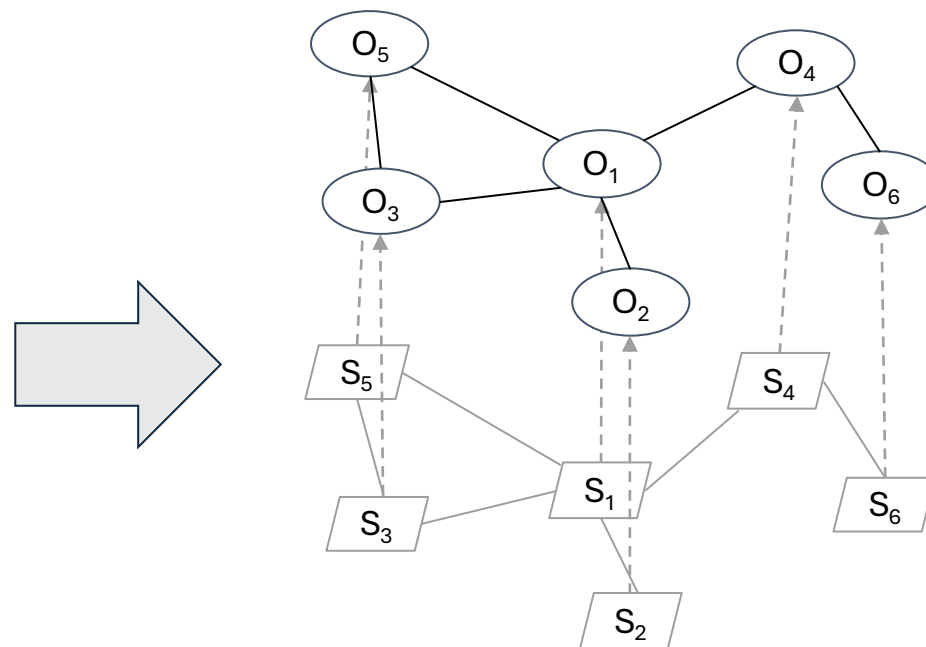
Sequences $\rightarrow$ 1D graphs

Approach 2: Hidden Markov Random Fields

Hidden Markov Model



Hidden Markov Random Field



Sequences $\rightarrow$ 1D graphs

Arbitrary graphs

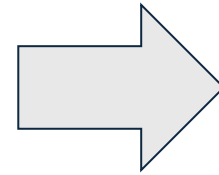
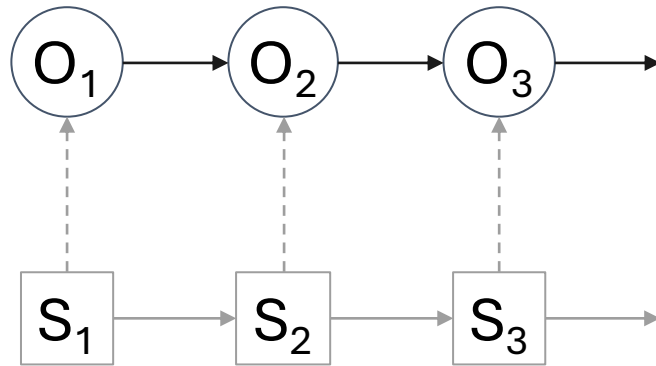
Zhao et al. "Spatial transcriptomics at subspot resolution with BayesSpace." *Nature Biotechnology* (2021).

Liu et al. "Joint dimension reduction and clustering analysis of single-cell RNA-seq and spatial transcriptomics data." *Nucleic acids research* (2022).

Liu et al. "Probabilistic embedding, clustering, and alignment for integrating spatial transcriptomics data with PRECAST." *Nature Communications* (2023).

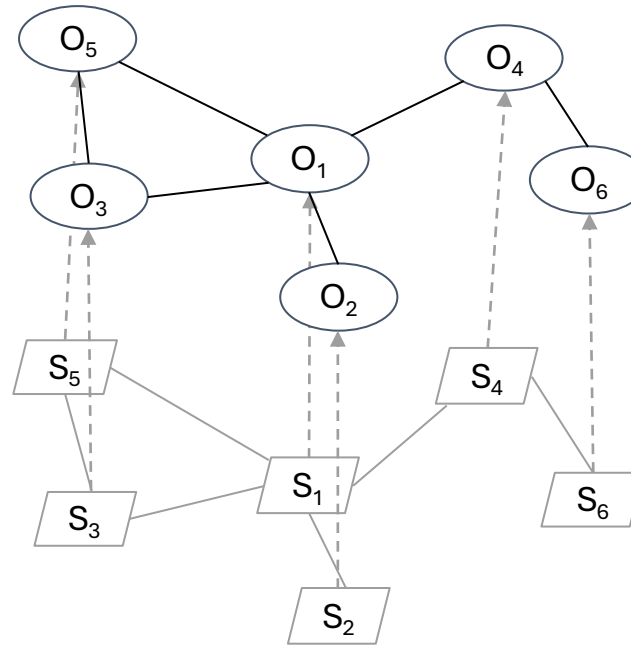
Approach 2: Hidden Markov Random Fields

Hidden Markov Model



Hidden Markov Random Field

Gene expression



Spatial domain

Sequences $\rightarrow$ 1D graphs

Arbitrary graphs

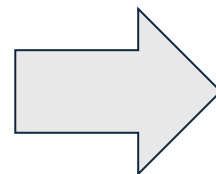
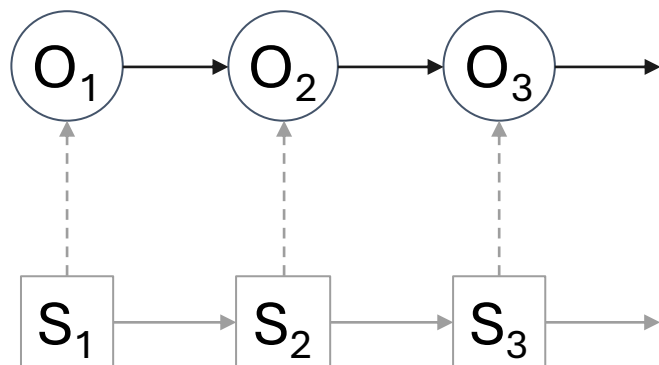
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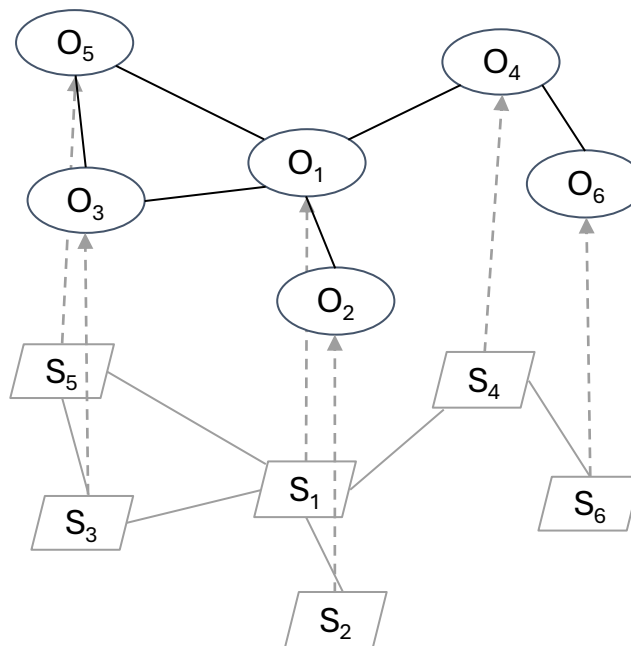
Approach 2: Hidden Markov Random Fields

Hidden Markov Model



Hidden Markov Random Field

Gene expression



Spatial domain

Arbitrary graphs

- Medium scalability
- No annotation required
Works directly on gene expression

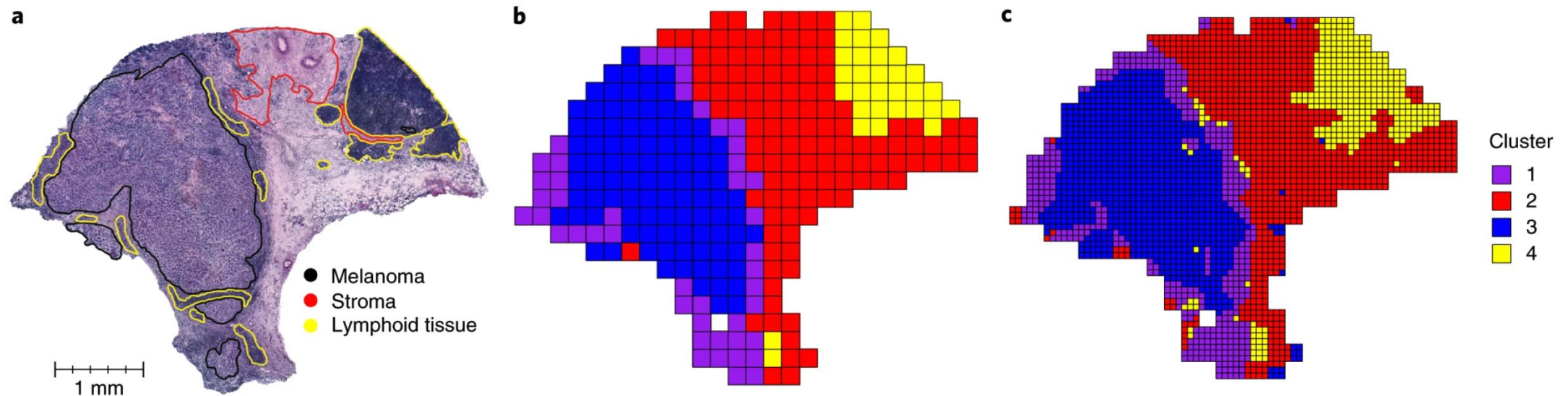
Sequences → 1D graphs

Zhao et al. "Spatial transcriptomics at subspot resolution with BayesSpace." *Nature Biotechnology* (2021).

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Approach 2: Hidden Markov Random Fields



Zhao et al. "Spatial transcriptomics at subspot resolution with BayesSpace." *Nature Biotechnology* (2021).

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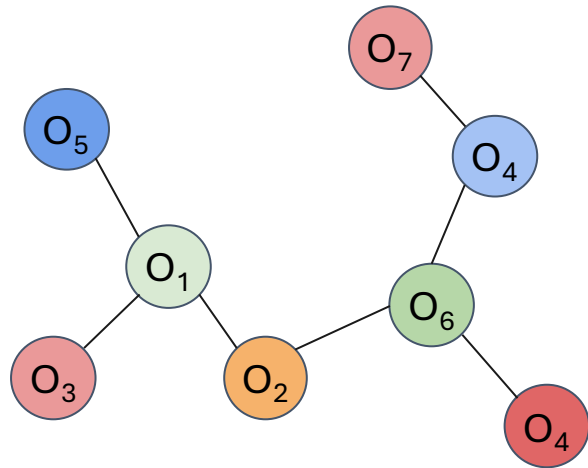
Liu et al. "Probabilistic embedding, clustering, and alignment for integrating spatial transcriptomics data with PRECAST." *Nature Communications* (2023).

Approach 3: Graph Neural Networks

Nodes with **similar neighborhoods** have **similar representations** (i.e. vectors)

Approach 3: Graph Neural Networks

Nodes with **similar neighborhoods** have **similar representations** (i.e. vectors)



  **Different cell types** → very different gene expression

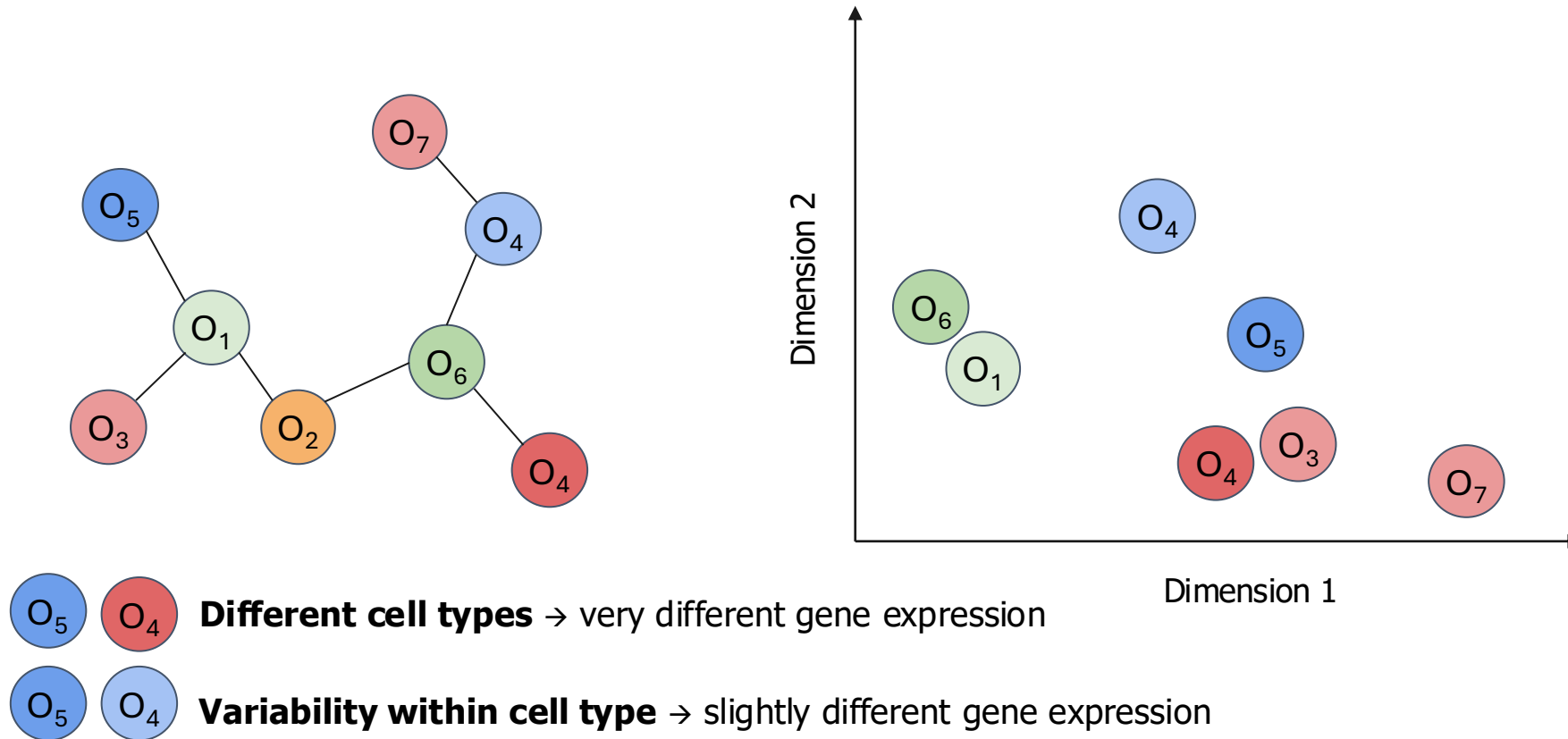
  **Variability within cell type** → slightly different gene expression

A Gentle Introduction to Graph Neural Networks

<https://distill.pub/2021/gnn-intro/>

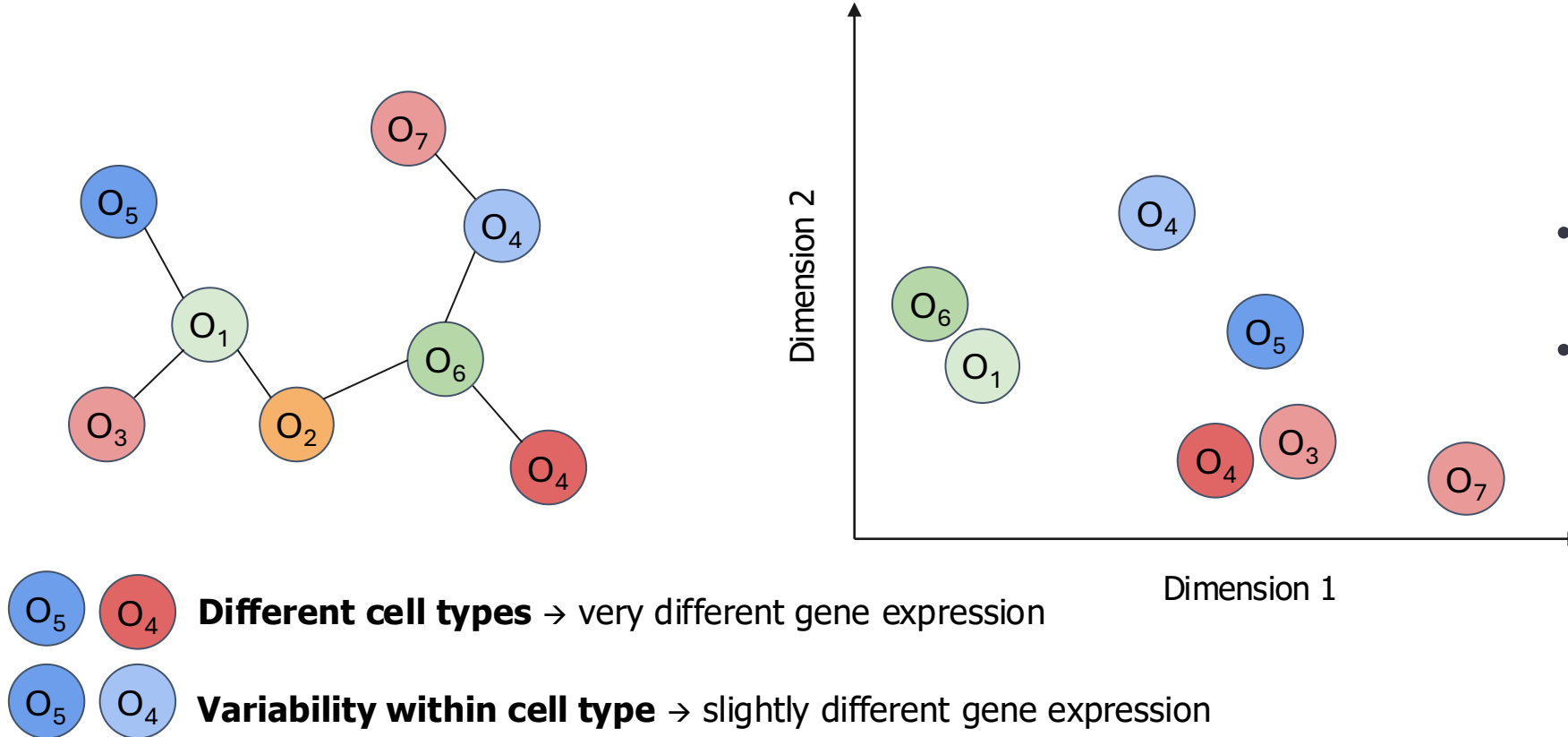
Approach 3: Graph Neural Networks

Nodes with **similar neighborhoods** have **similar representations** (i.e. vectors)



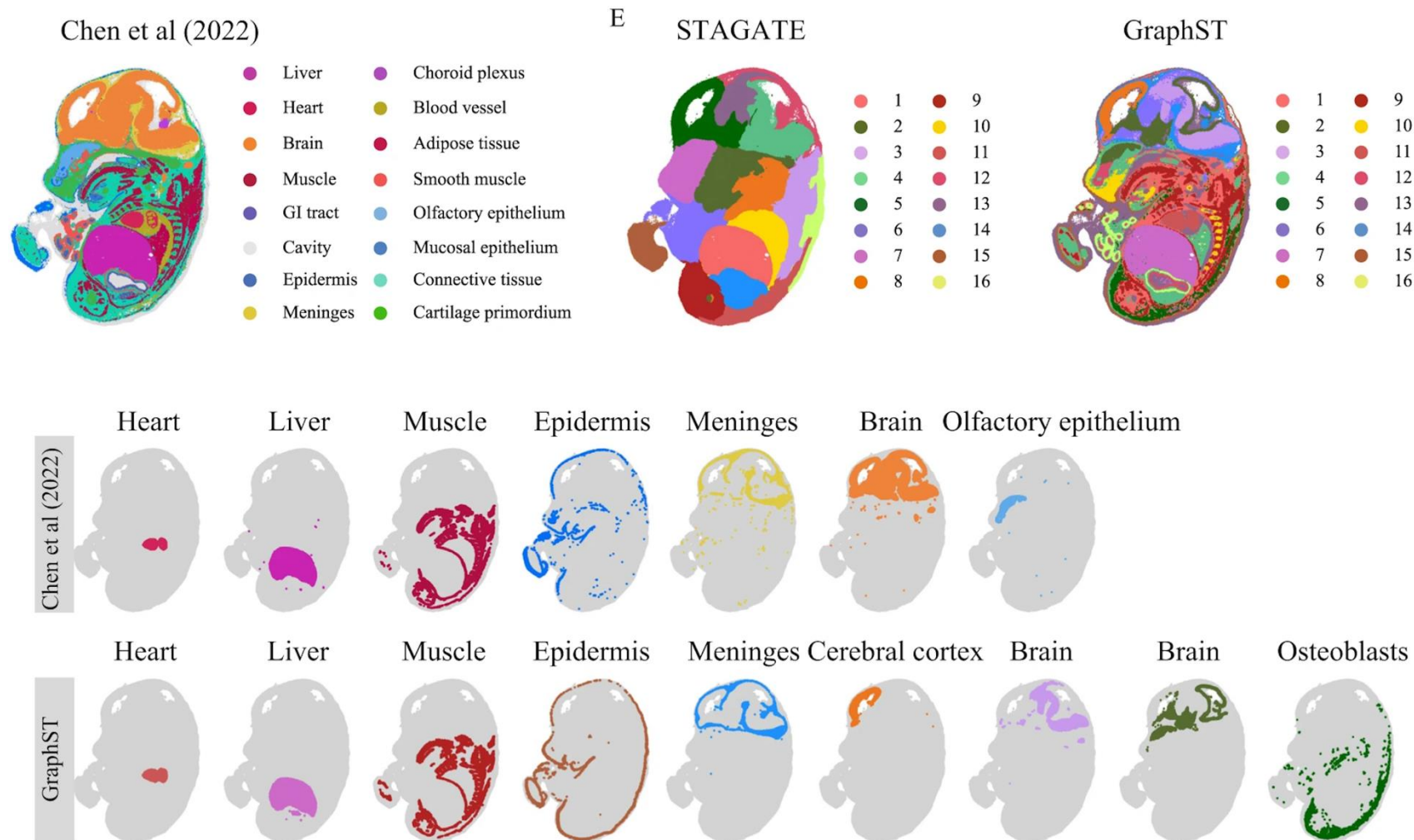
Approach 3: Graph Neural Networks

Nodes with **similar neighborhoods** have **similar representations** (i.e. vectors)

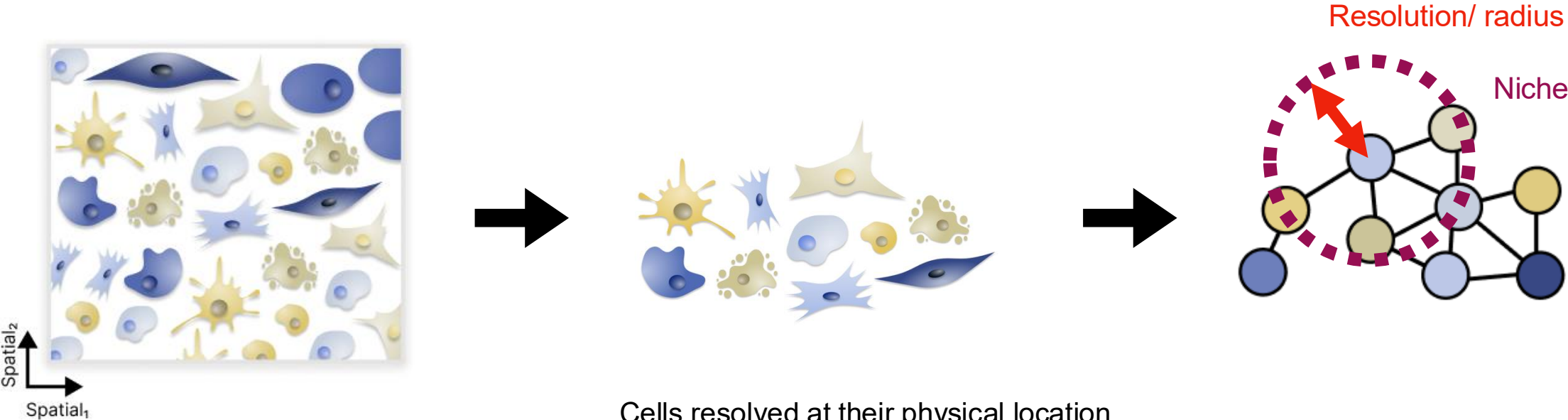


- Low/medium scalability
- No annotation required
Works directly on gene expression

Approach 3: Graph Neural Networks



Defining spatial graphs of cells based on spatial proximity

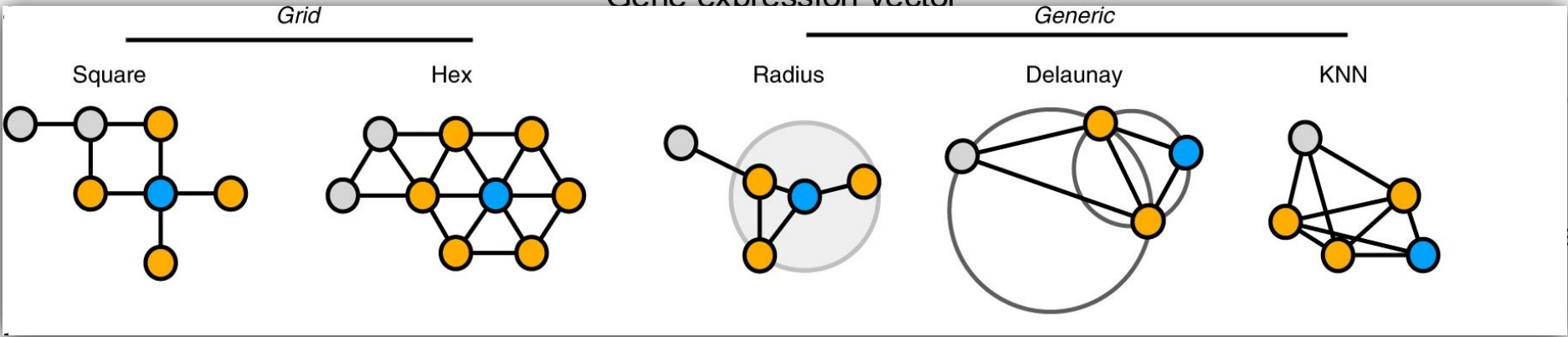


Segmented targeted spatial omics

Cells resolved at their physical location

- Coordinates
- Gene expression vector

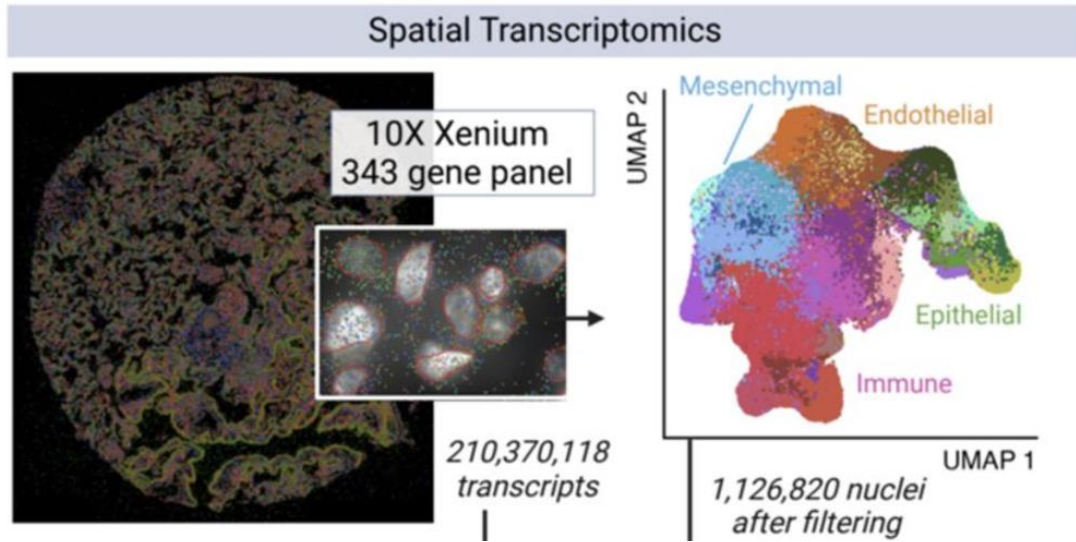
Spatial graph of cells



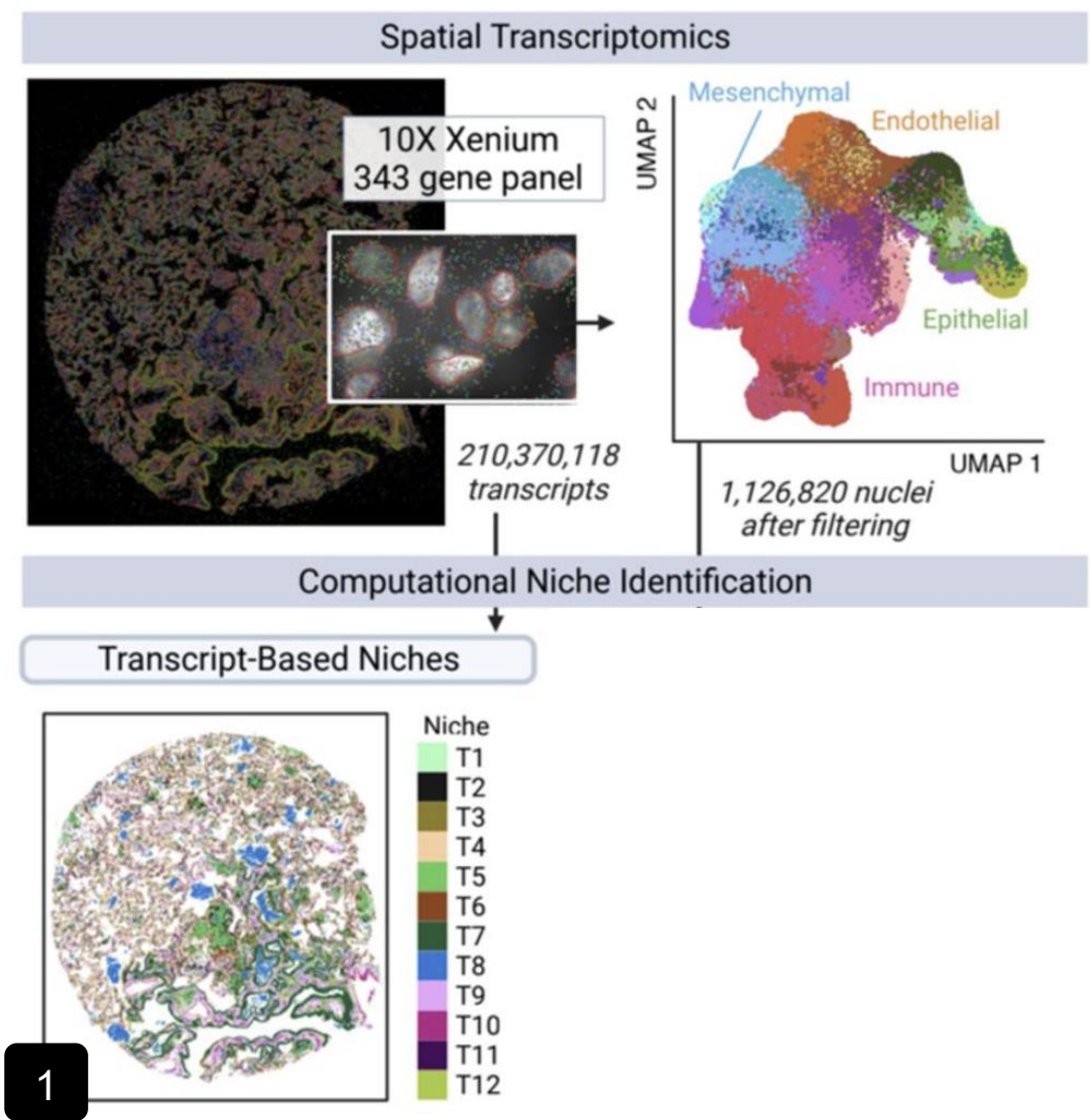
Proximity matrix

$d(i, j) \leq r$ with d being the distance between two

How to define niche labels in practice?



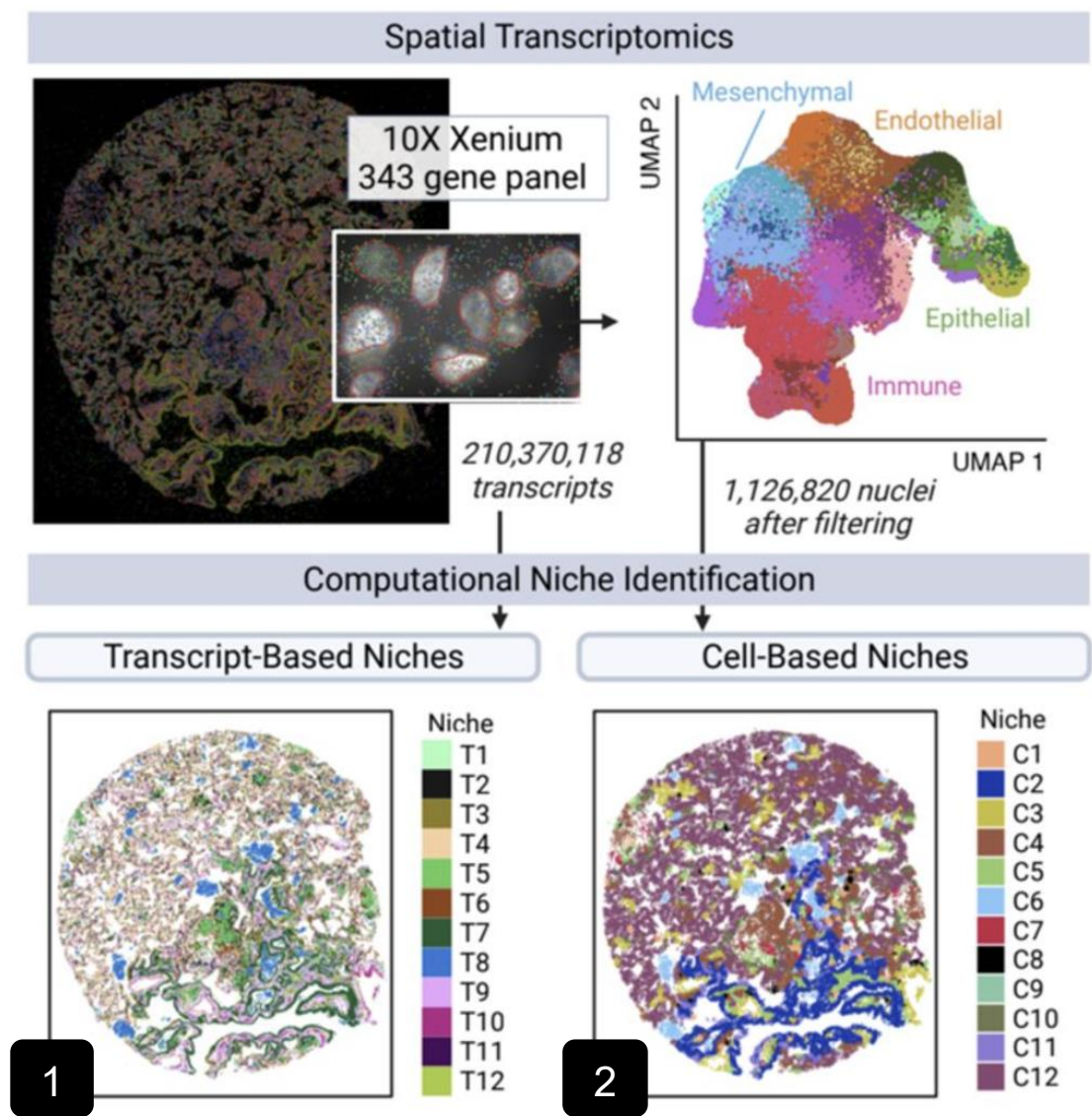
How to define niche labels in practice?



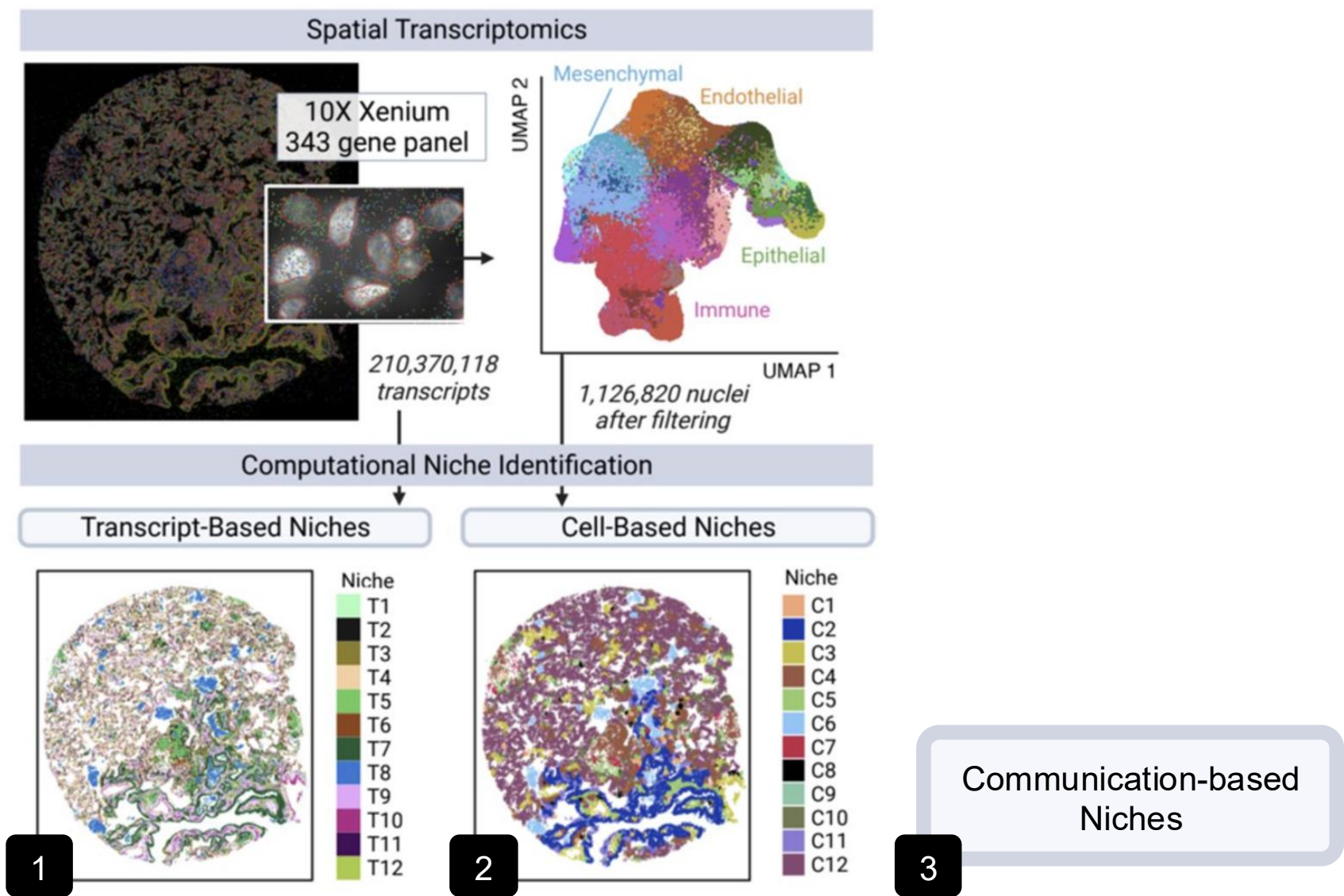
1

Vannan, A., *et al.* Image-based spatial transcriptomics identifies molecular niche dysregulation associated with distal lung remodeling in pulmonary fibrosis. *bioRxiv*, (2023).

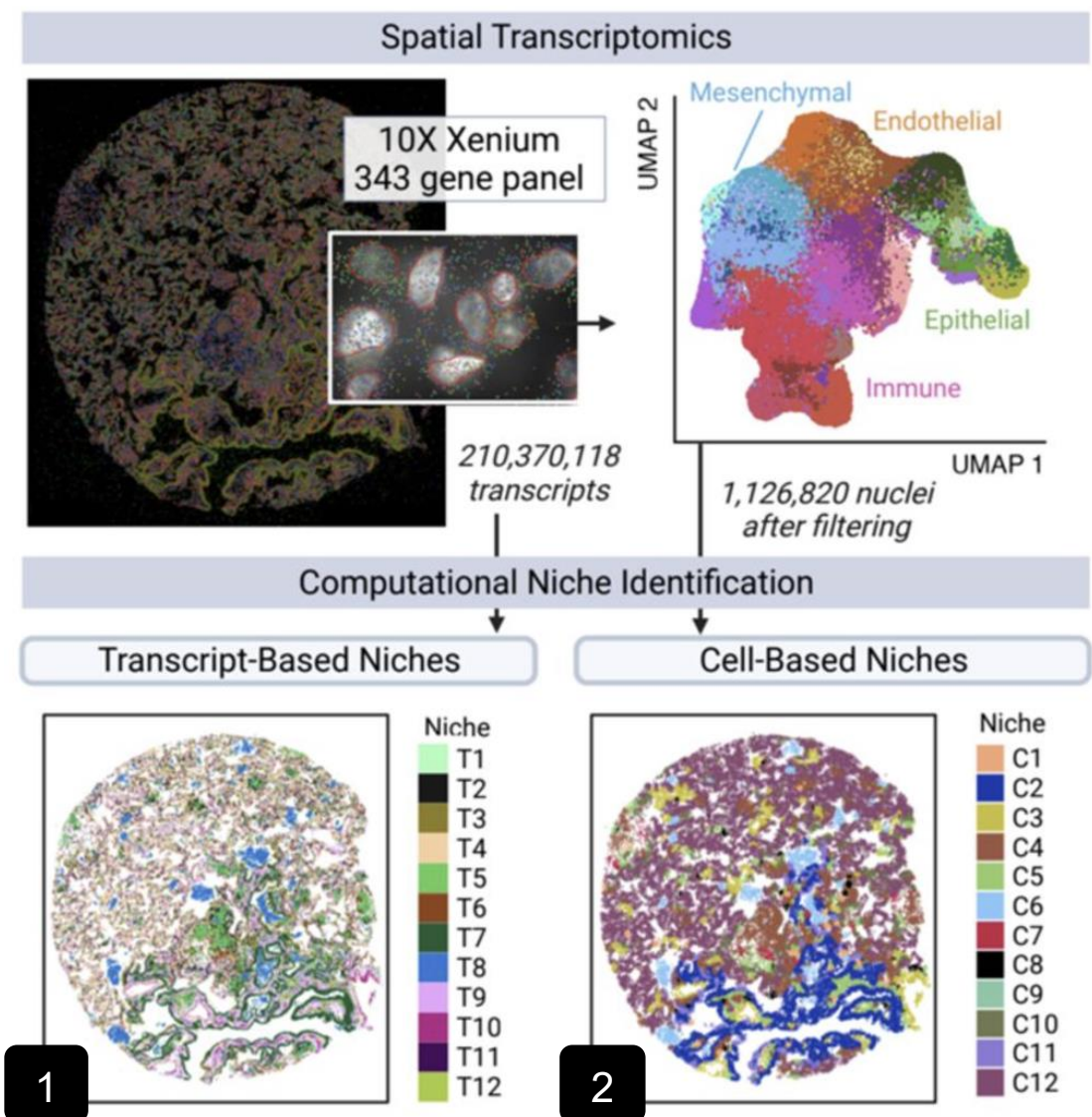
How to define niche labels in practice?



How to define niche labels in practice?



How to define niche labels in practice?



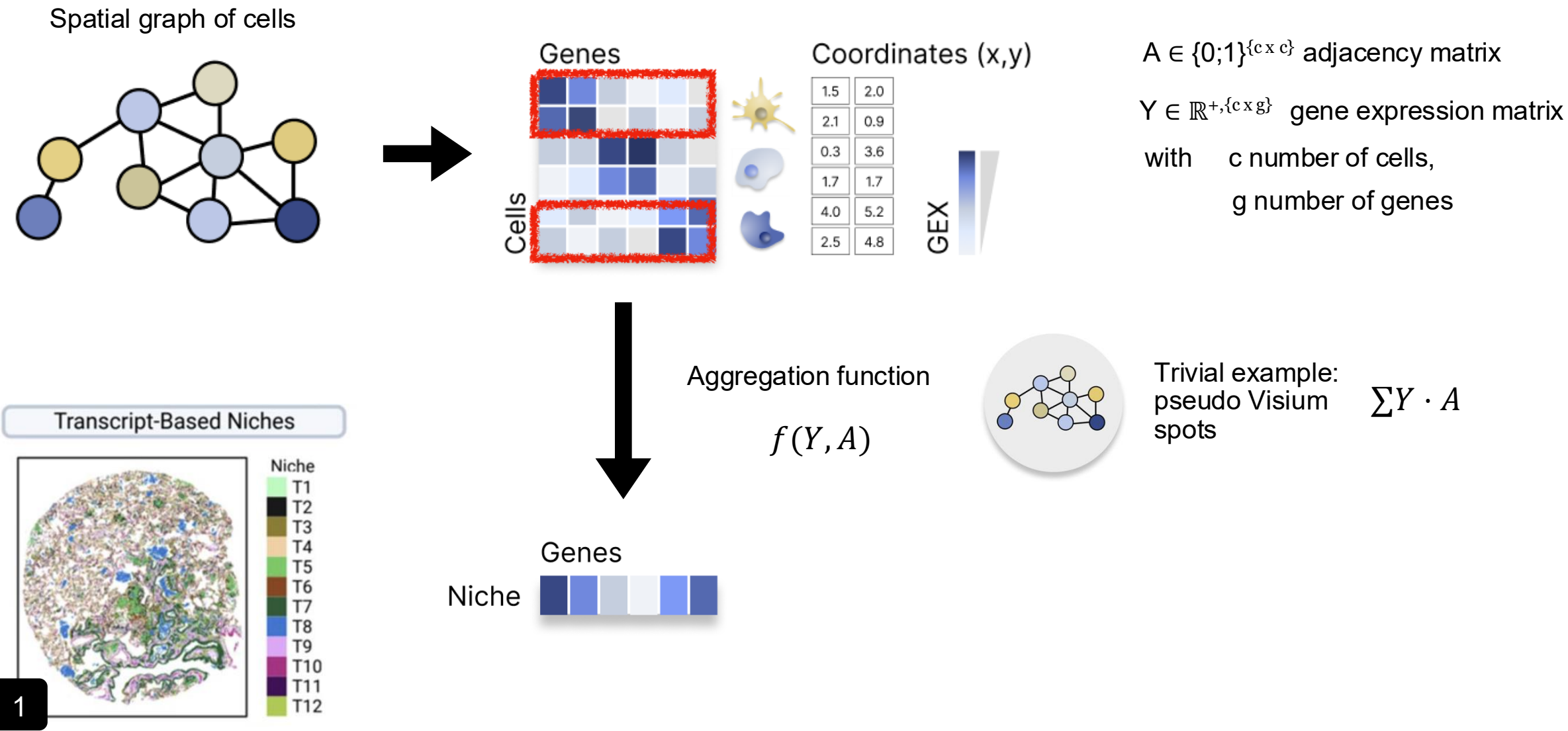
Note:

- Niche labels can be very tissue dependent
- Niche labels strongly depend on the specific use-case you are interested in
- Batch-effects might not be only technical variability, but biological differences

Communication-based Niches

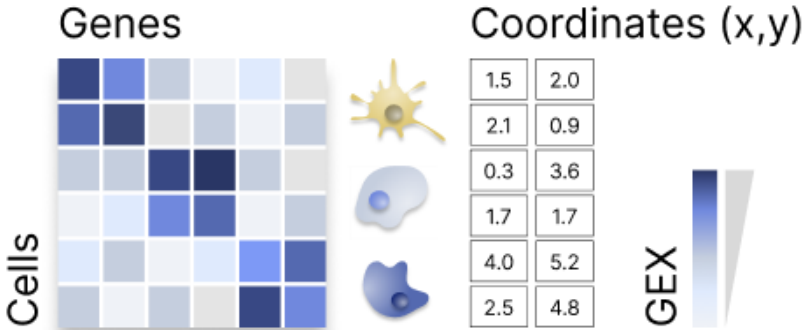
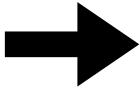
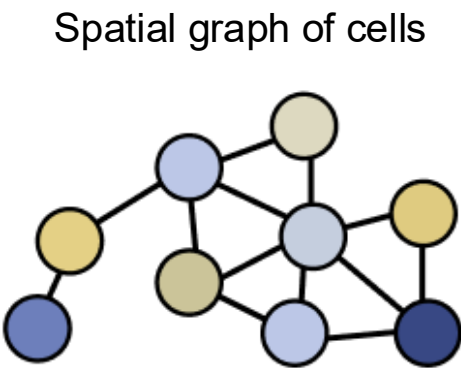
Transcription-based niches

i.e. define a niche based on the expression profile of its surrounding neighbours



Transcription-based niches

i.e. define a niche based on the expression profile of its surrounding neighbours



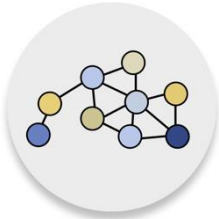
$A \in \{0;1\}^{\{c \times c\}}$ adjacency matrix

$Y \in \mathbb{R}^{+, \{c \times g\}}$ gene expression matrix

with c number of cells,
 g number of genes

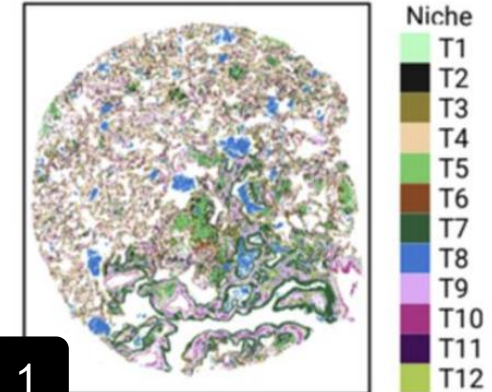
Aggregation function

$f(Y, A)$



Trivial example:
pseudo Visium spots $\sum Y \cdot A$

Transcript-Based Niches



Article | [Open access](#) | Published: 27 February 2024

BANKSY unifies cell typing and tissue domain segmentation for scalable spatial omics data analysis

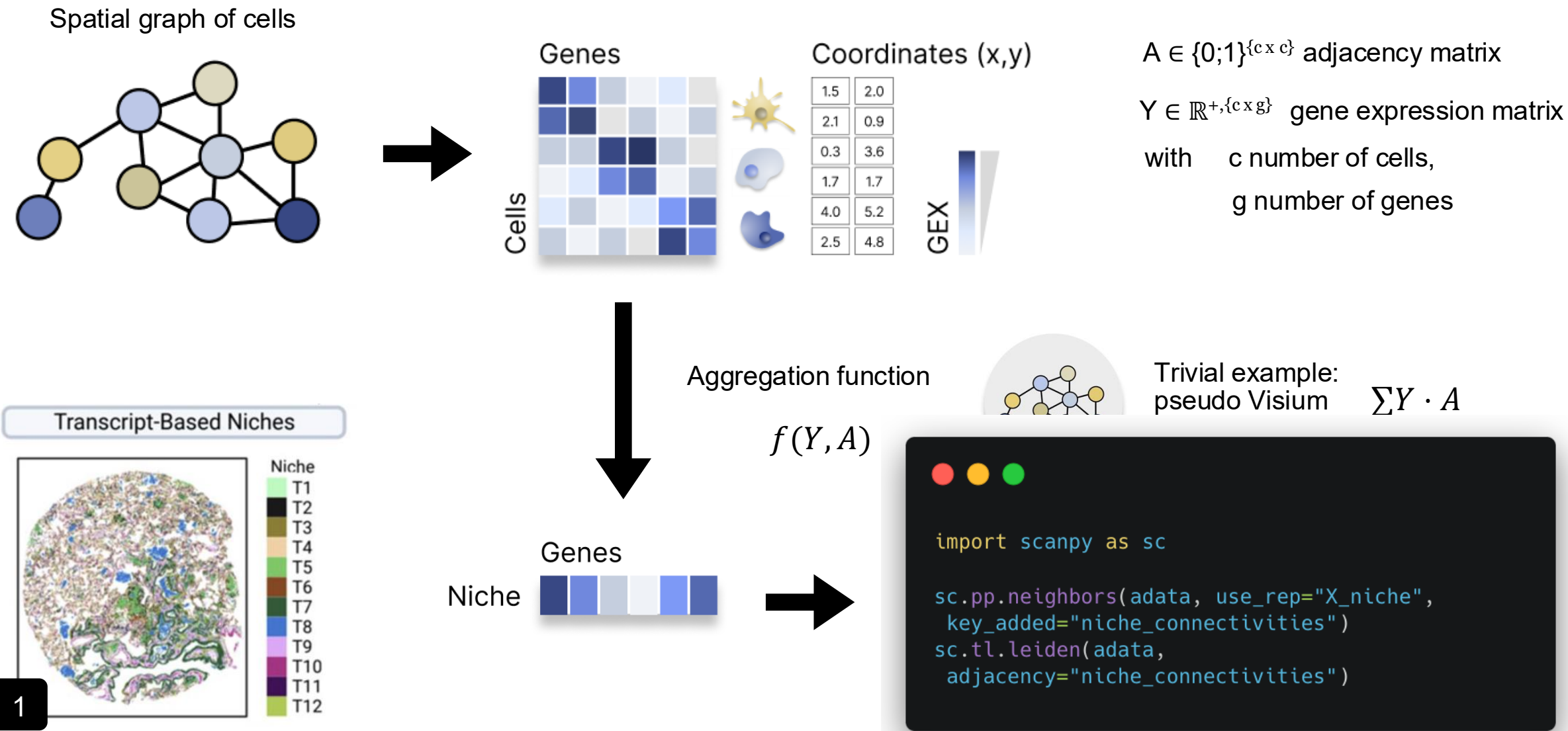
[Vipul Singhal](#), [Nigel Chou](#), [Joseph Lee](#), [Yifei Yue](#), [Jinyue Liu](#), [Wan Kee Chock](#), [Li Lin](#), [Yun-Ching Chang](#), [Erica Mei Ling Teo](#), [Jonathan Aow](#), [Hwee Kuan Lee](#), [Kok Hao Chen](#) & [Shyam Prabhakar](#)

Nature Genetics **56**, 431–441 (2024) | [Cite this article](#)



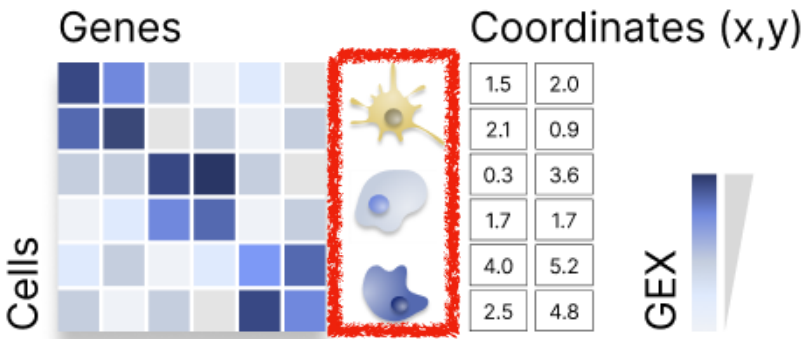
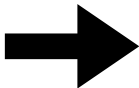
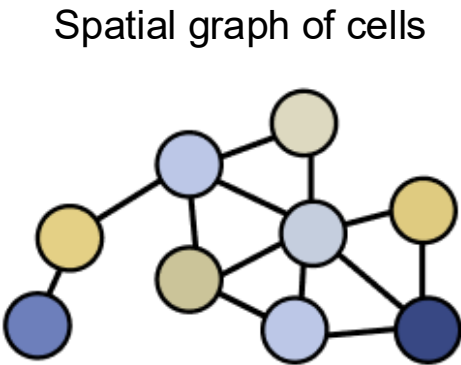
Transcription-based niches

i.e. define a niche based on the expression profile of its surrounding neighbours



Cell-based niches

i.e. define a niche based on the composition of neighbouring cell types



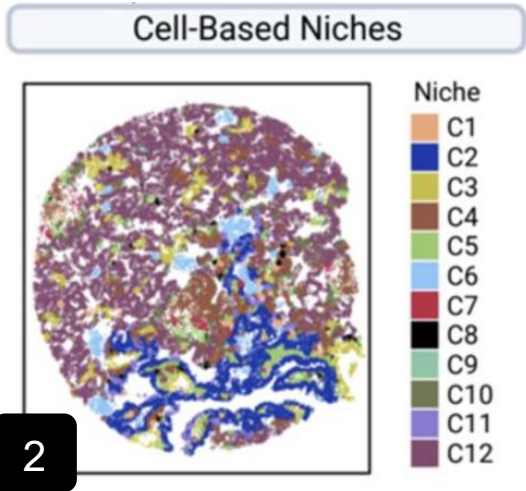
$A \in \{0;1\}^{\{c \times c\}}$ adjacency matrix

$X \in \{0,1\}^{\{c \times d\}}$ one-hot encoded cell-type matrix

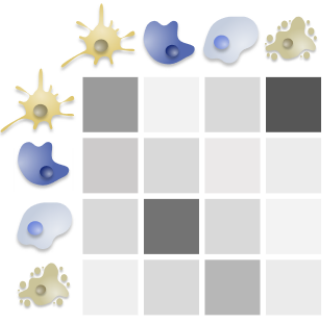
with c number of cells,
 d number of cell-types

Aggregation function

$$f(X, A)$$



Cell-type composition of the niche



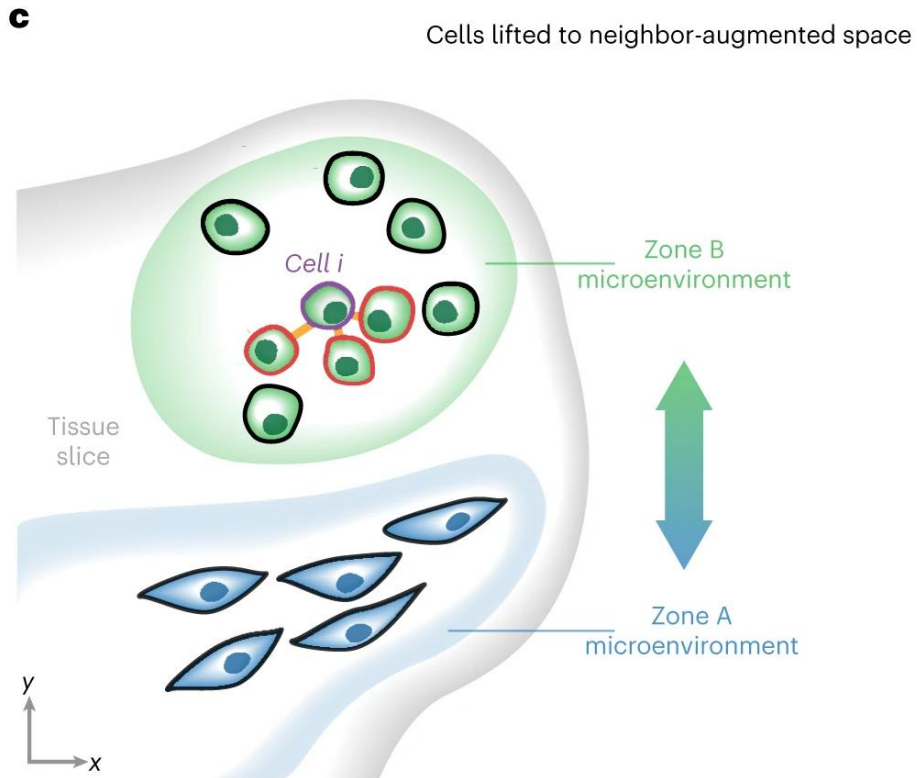
```
import scanpy as sc

sc.pp.neighbors(adata, use_rep="X_niche",
key_added="niche_connectivities")
sc.tl.leiden(adata,
adjacency="niche_connectivities")
```

Examples of domain identification methods

BANKSY

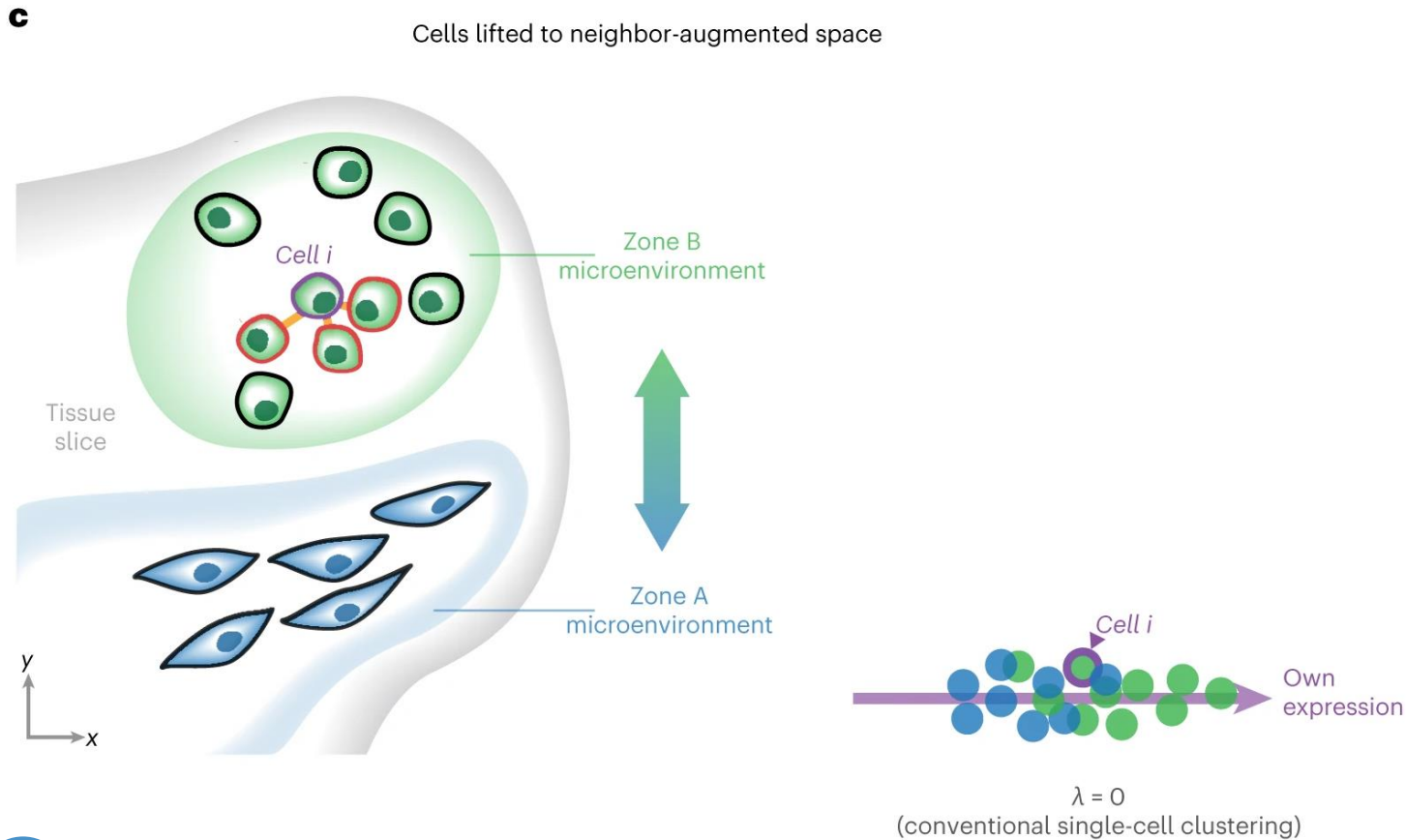
Assumption: a cell's state can be more fully represented by considering both its own transcriptome and that of its local microenvironment.



● and ● have “similar” gene expression

BANKSY

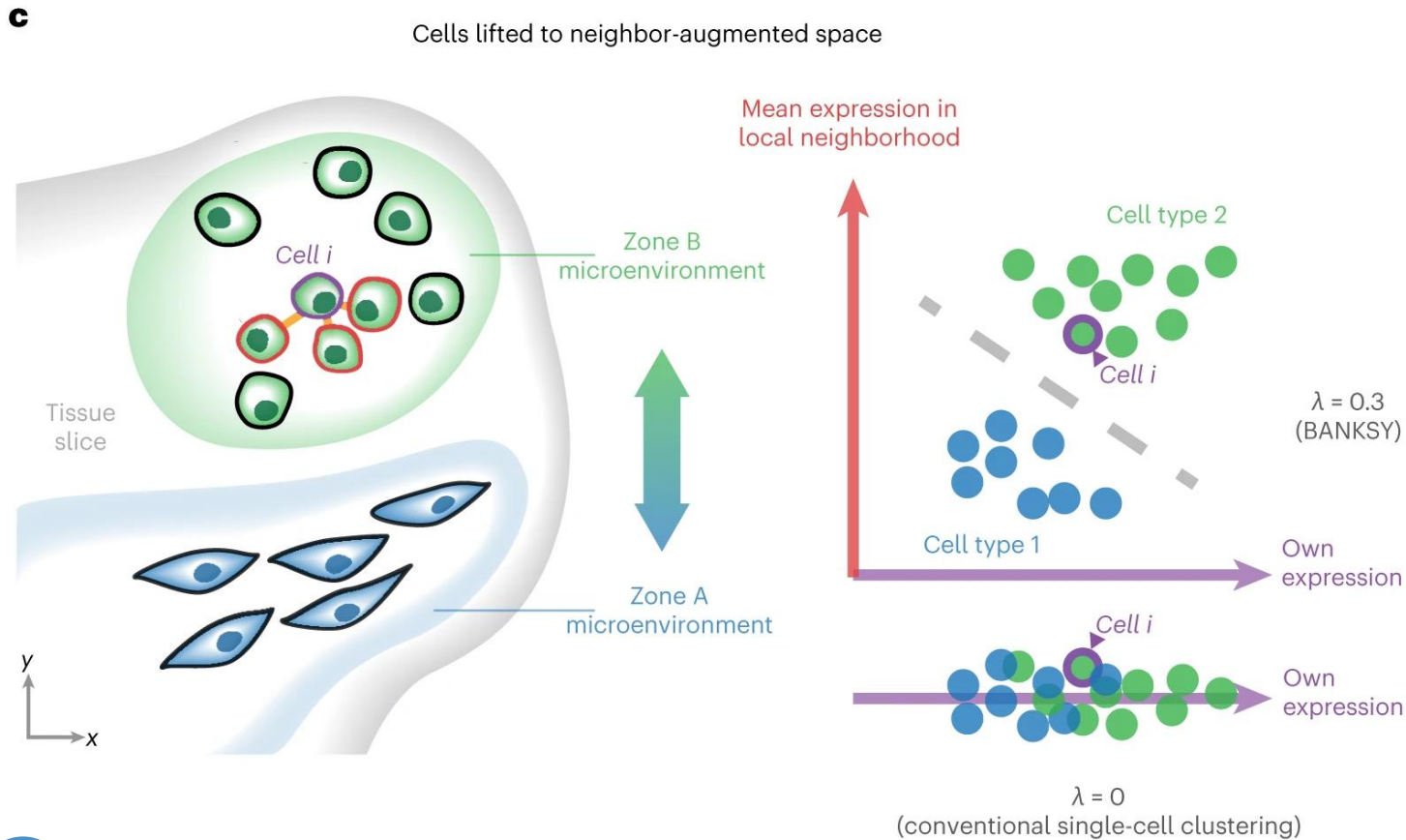
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● and ● have “similar” gene expression

BANKSY

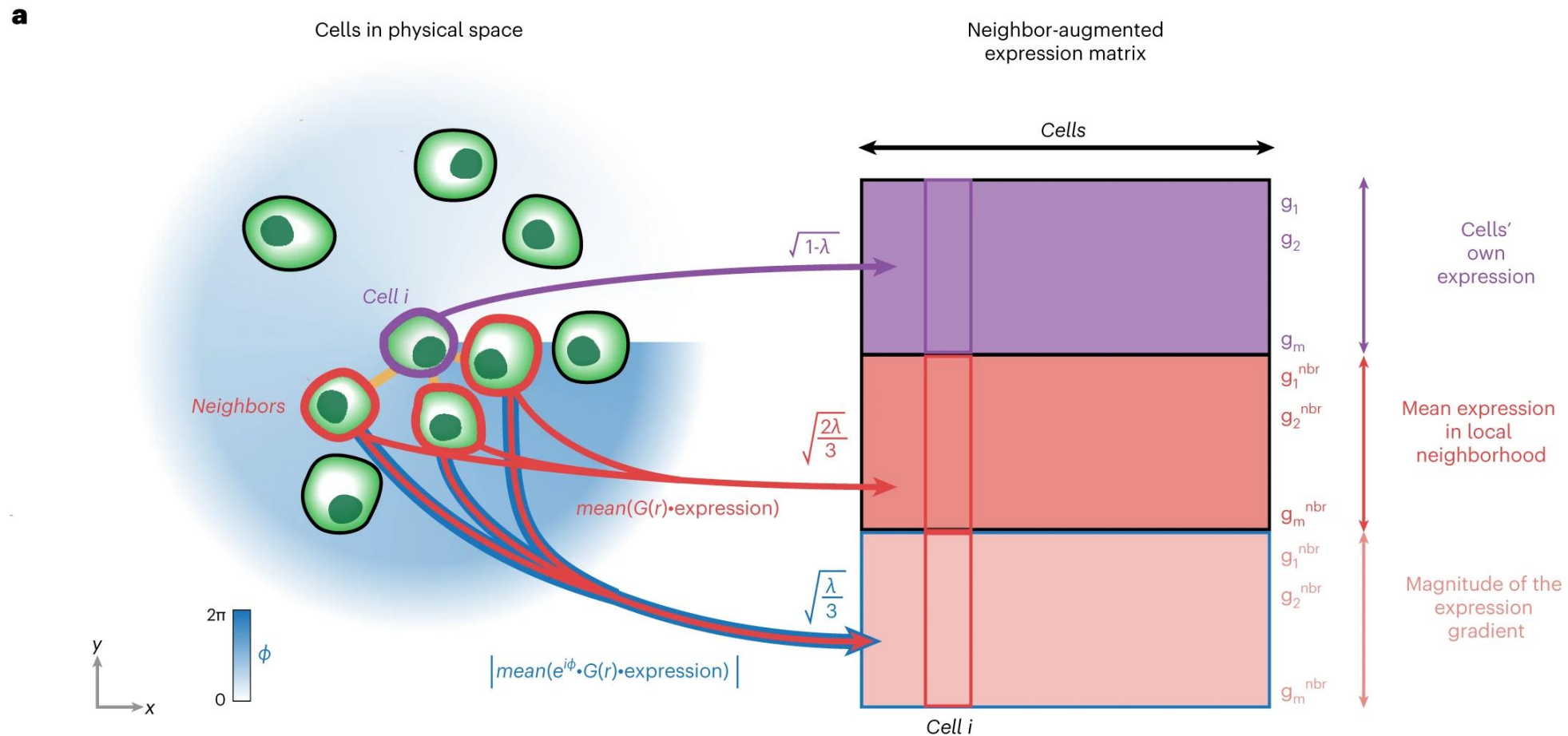
Assumption: a cell's state can be more fully represented by considering both its own transcriptome and that of its local microenvironment.



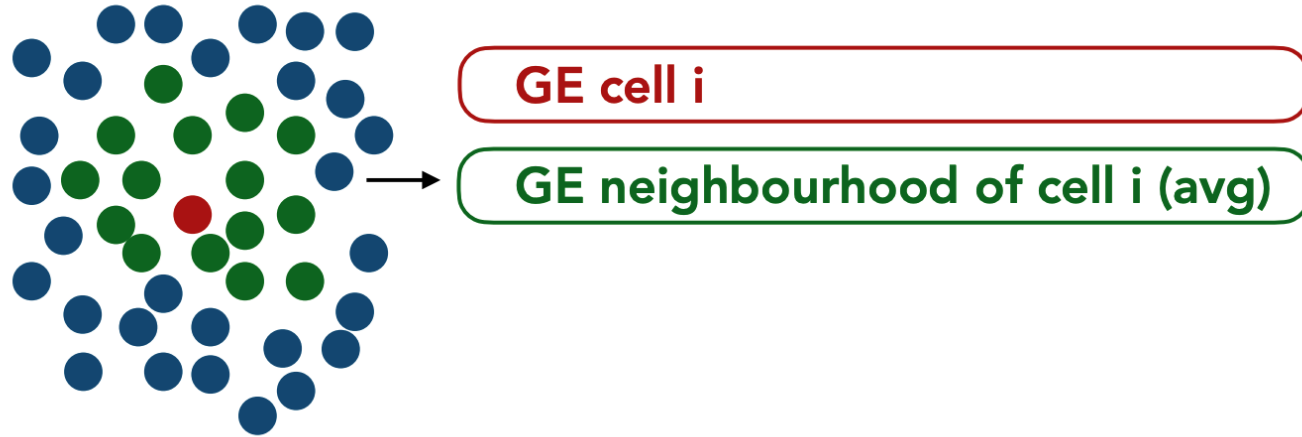
● and ● have “similar” gene expression

BANKSY

Assumption: a cell's state can be more fully represented by considering both its own transcriptome and that of its local microenvironment.



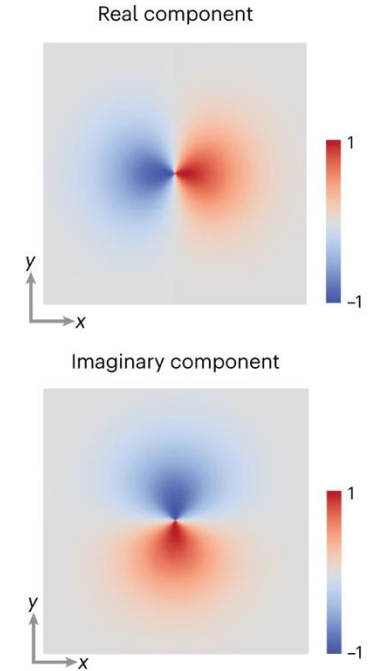
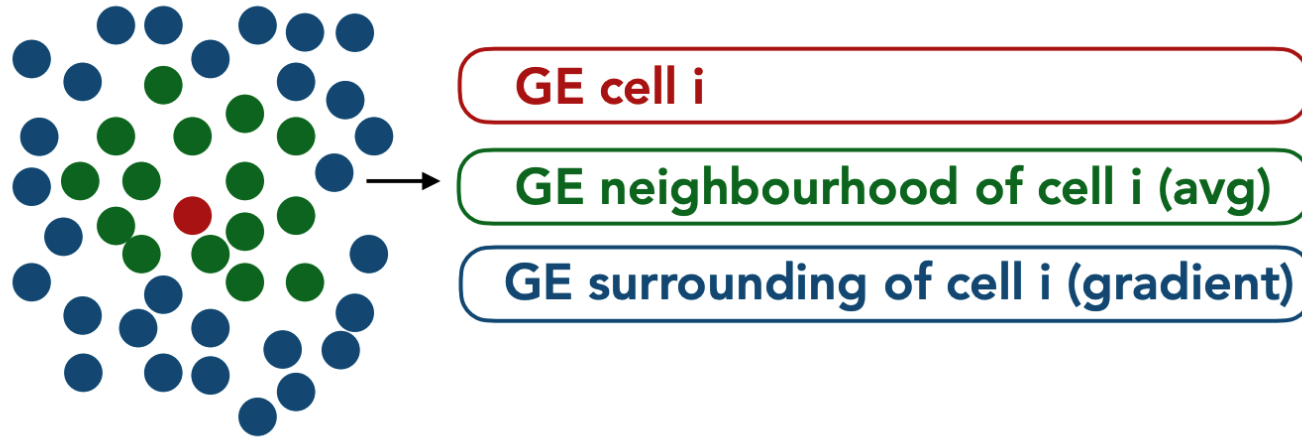
BANKSY



From the original count matrix (matrix C):

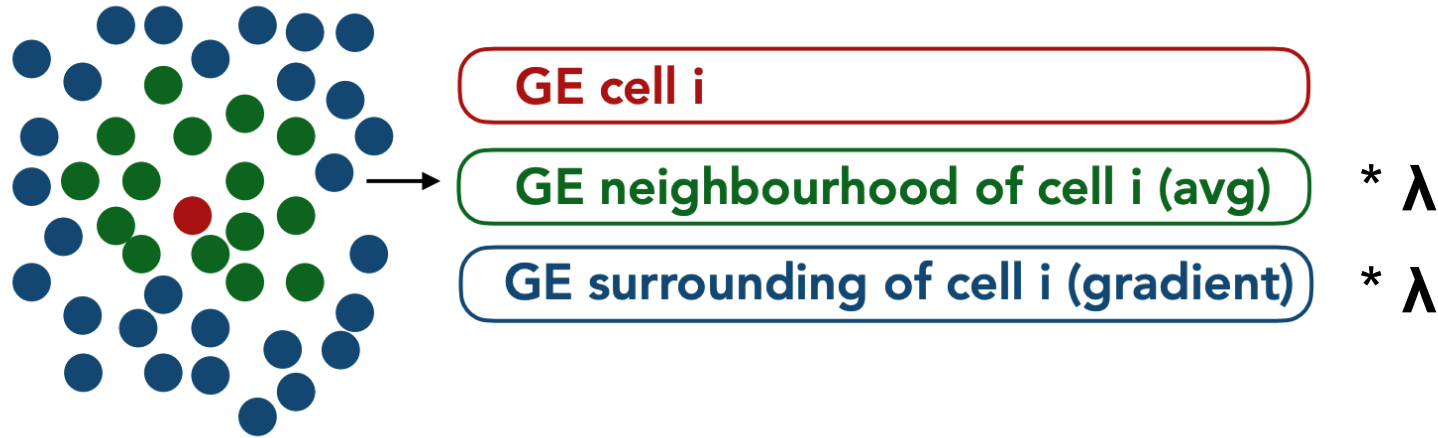
- Construct a **neighborhood graph** between cells in physical space using k-nearest neighbors (Matrix M) and compute the mean expression in the neighborhood of a cell

BANKSY



From the original count matrix (matrix C):

- Construct a **neighborhood graph** between cells in physical space using k -nearest neighbors (Matrix M) and compute the mean expression in the neighborhood of a cell
- AGF (Azimuthal Gabor Filter)**: measures gradients in gene expression in each neighborhood (size= $2*k$). (Matrix G)



From the original count matrix (matrix **C**):

- Construct a **neighborhood graph** between cells in physical space using k-nearest neighbors (Matrix **M**) and compute the mean expression in the neighborhood of a cell
- AGF**: measures gradients in gene expression in each neighborhood (size= $2*k$). (Matrix **G**)
- Matrices **M** and **G** are **scaled** based on a mixing parameter λ , which controls their relative weighting.
- Concatenate** matrices **M**, **G** and **C**.

BANKSY

Step by step

1. Normalise data
2. Construction of k-nearest-neighbors based on physical distance
3. Generate spatial weights from distance
4. Generate Banksy matrix = From the original count matrix (matrix C):
 - a. Construct a **neighborhood graph** between cells in physical space using k-nearest neighbors (Matrix **M**) and compute the mean expression in the neighborhood of a cell
 - b. AGF**: measures gradients in gene expression in each neighborhood. (Matrix **G**)
 - c. Matrices M and G are scaled based on a mixing parameter λ , which controls their relative weighting.
 - d. Concatenate** matrices M, G and C.
6. Dimensionality reduction (PCA)
7. Run clustering (Leiden)

BANKSY

Important parameters:

- Weighting parameter $\lambda \in [0, 1]$

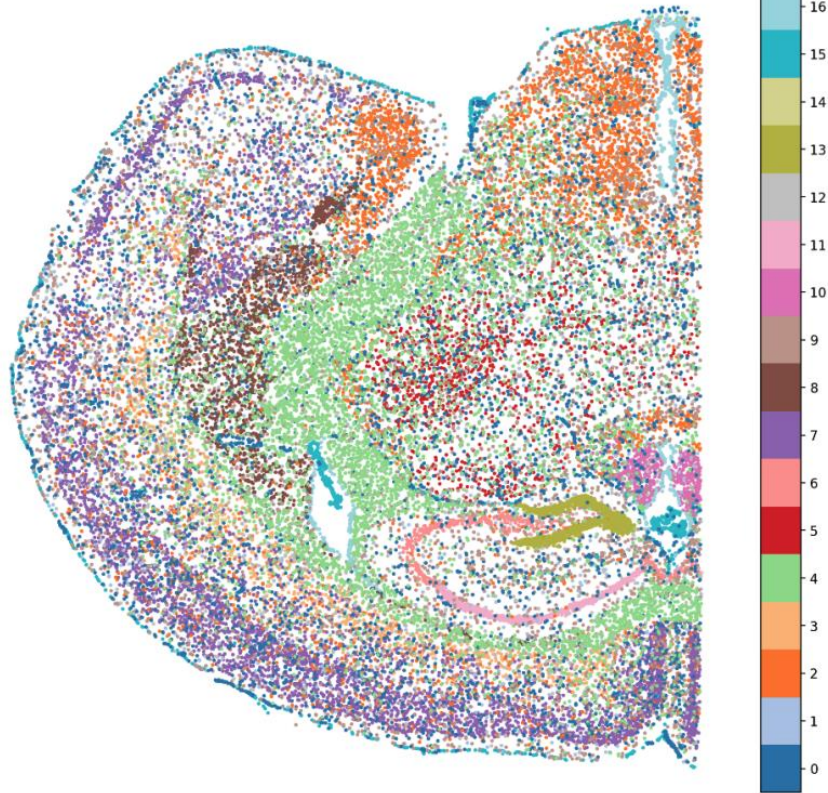
Weights the **contributions** of the cell-transcriptome matrix and the neighbor expression matrices. **Smaller λ settings emphasize cells' own transcriptomes** (and cause cells to cluster according to cell type). In the limit, when $\lambda = 0$, BANKSY reduces to conventional nonspatial clustering.
- K-geom

Number of nearest neighbors to be included in the neighbourhood
 $2 * k\text{-geom} = \text{number of spatial units that will be calculated with the gradient}$
- Number of Principal Components
- Clustering resolution

BANKSY

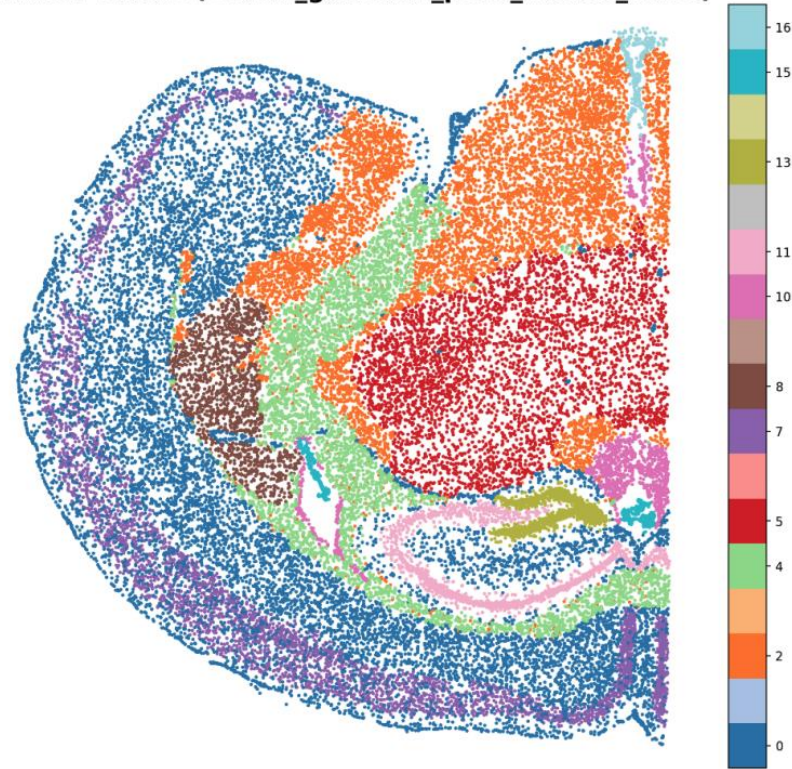
Lambda = 0

BANKSY Labels (nonspatial_pc20_nc0.00_r0.10)



Lambda = 0.6

BANKSY Labels (scaled_gaussian_pc20_nc0.60_r0.10)



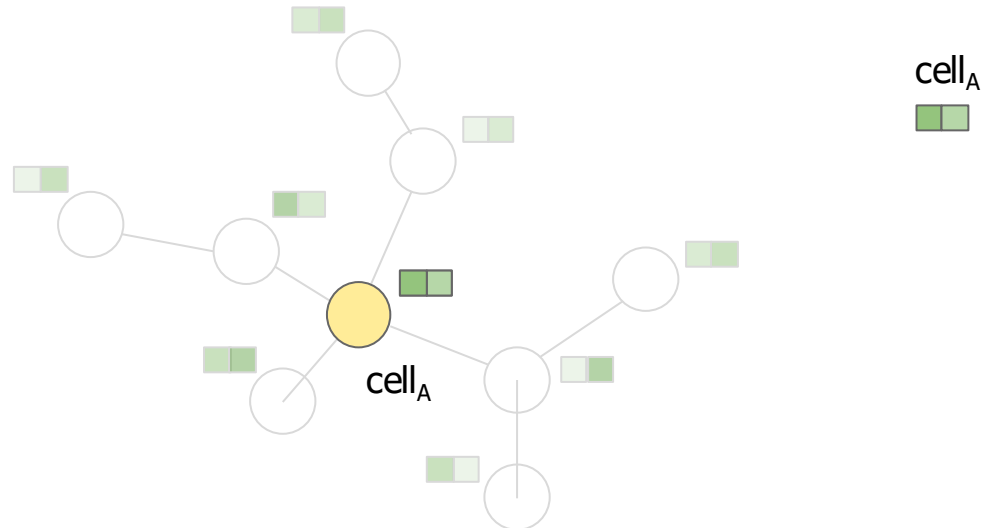
What will the spatial domains look like if $\lambda = 1$?

What if $k\text{-geom} = 100$?

CellCharter

Step by step

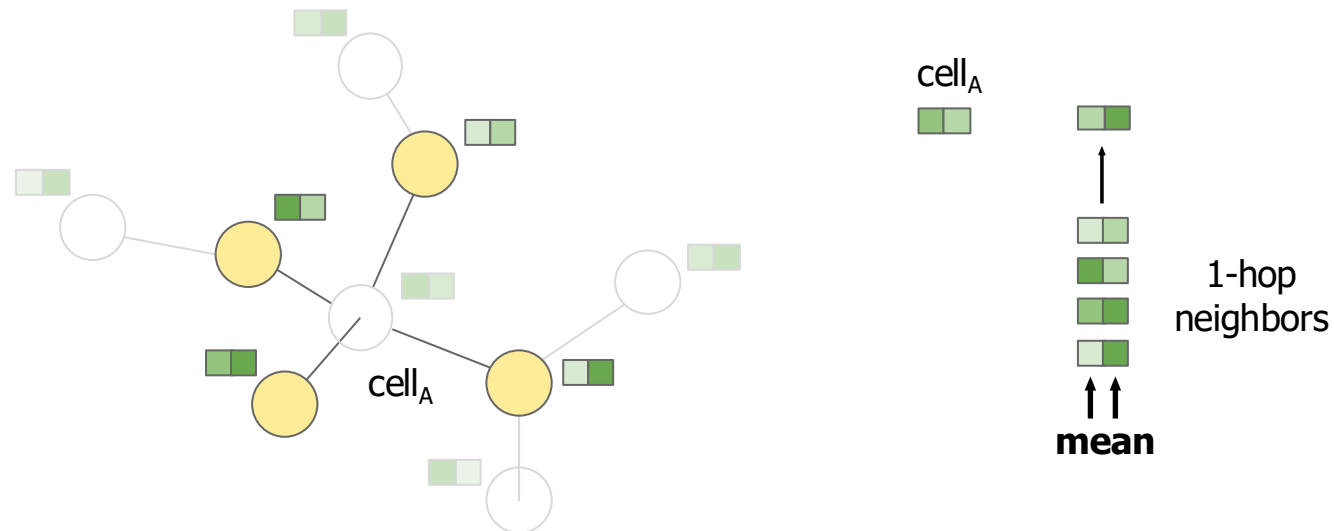
1. Dimensionality reduction and batch effect correction with VAE
2. Network based on proximity = l steps away * **with some checks for irregular structures**
 - a. Gene expression of cell A



CellCharter

Step by step

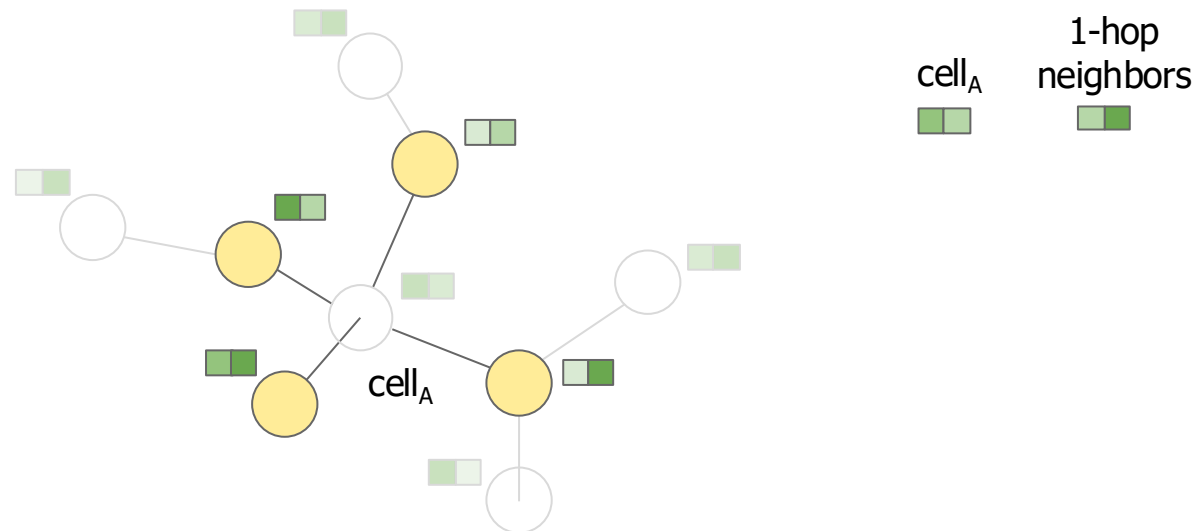
1. Dimensionality reduction and batch effect correction with VAE
2. Network based on proximity = 1 steps away
 - a. Gene expression of cell A
 - b. Mean gene expression of first neighborhood



CellCharter

Step by step

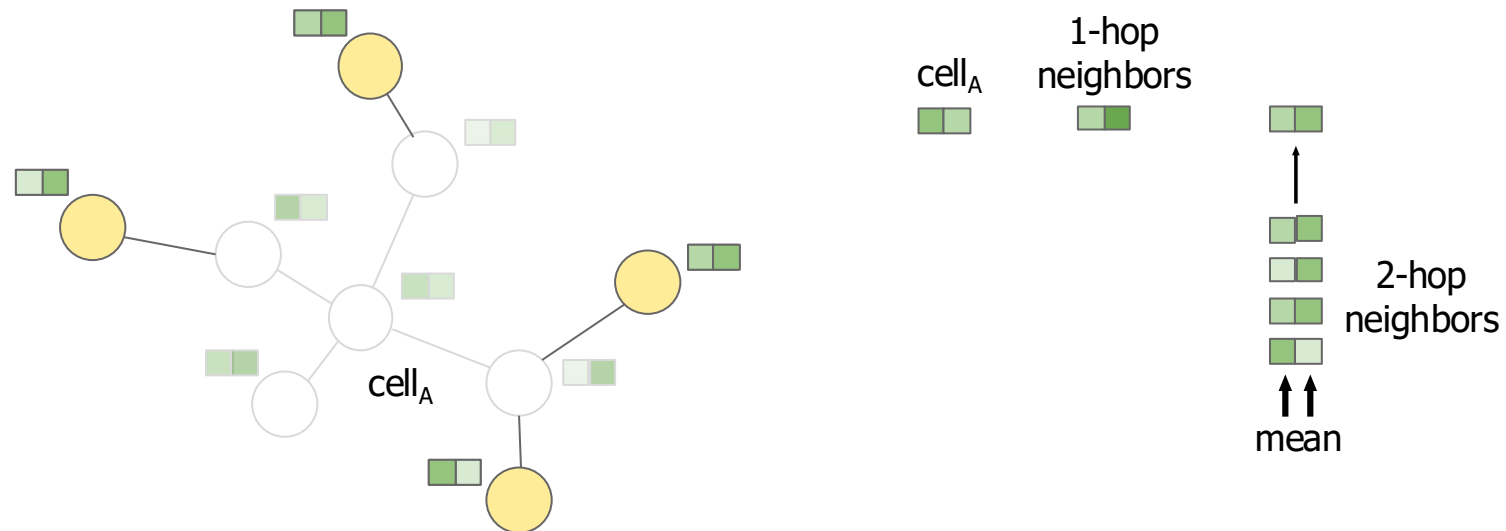
1. Dimensionality reduction and batch effect correction with VAE
2. Network based on proximity = l steps away
 - a. Gene expression of cell A
 - b. Mean gene expression of first neighborhood



CellCharter

Step by step

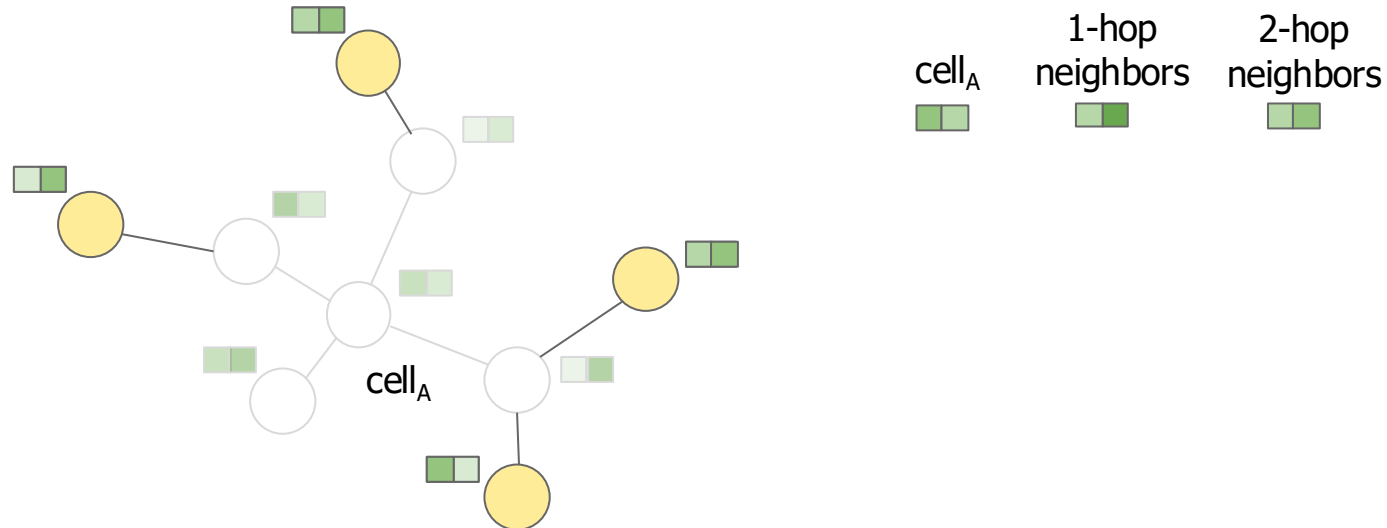
1. Dimensionality reduction and batch effect correction with VAE
2. Network based on proximity = 1 steps away
 - a. Gene expression of cell A
 - b. Mean gene expression of first neighbourhood
 - c. Mean gene expression of second neighbourhood



CellCharter

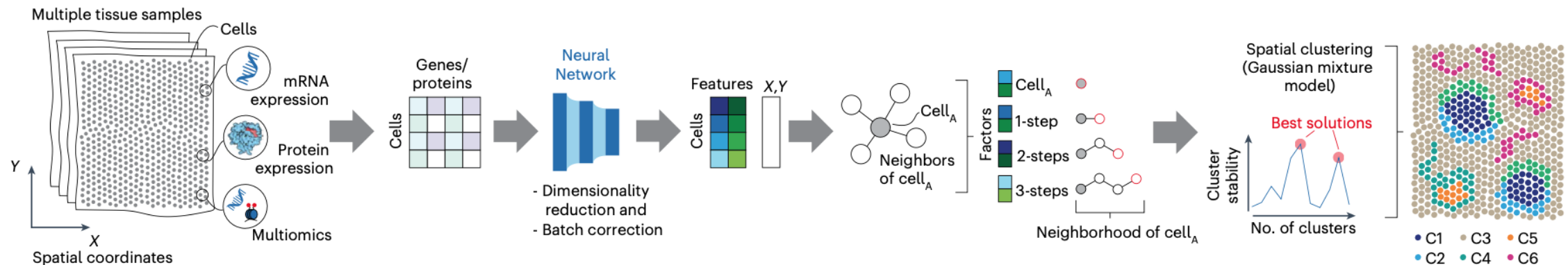
Step by step

1. Dimensionality reduction and batch effect correction with VAE
2. Network based on proximity = l steps away
 - a. Gene expression of cell A
 - b. Mean gene expression of first neighbourhood
 - c. Mean gene expression of second neighbourhood
 - d.
 - e. Mean gene expression of l -th neighbourhood

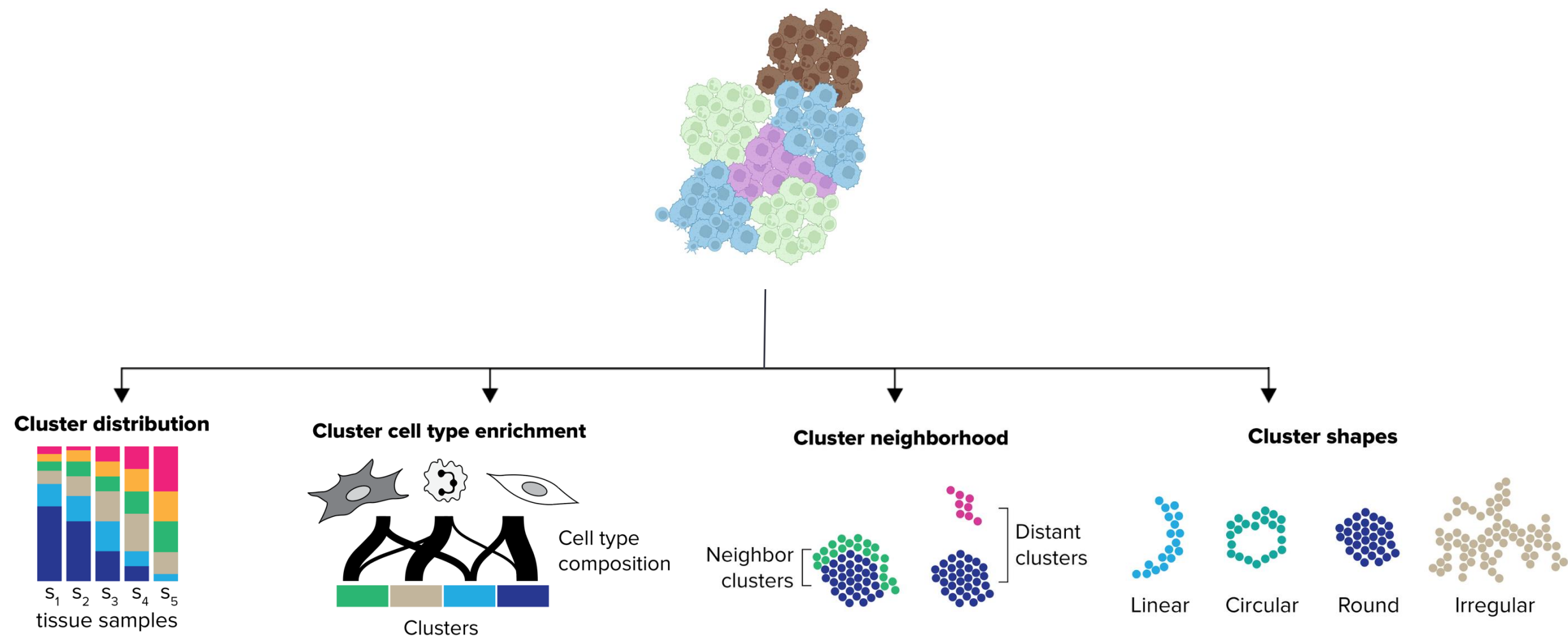


CellCharter

1. Dimensionality reduction and batch effect correction with VAE
2. Network based on proximity = l steps away
 - a. Gene expression of cell A
 - b. Mean gene expression of first neighbourhood
 - c. Mean gene expression of second neighbourhood
 - d.
 - e. Mean gene expression of l -th neighbourhood
3. Concatenation (neighbourhood aggregation method to obtain one feature vector from the concatenation of multiple feature vectors)
4. Clustering using a Gaussian mixture model (GMM)



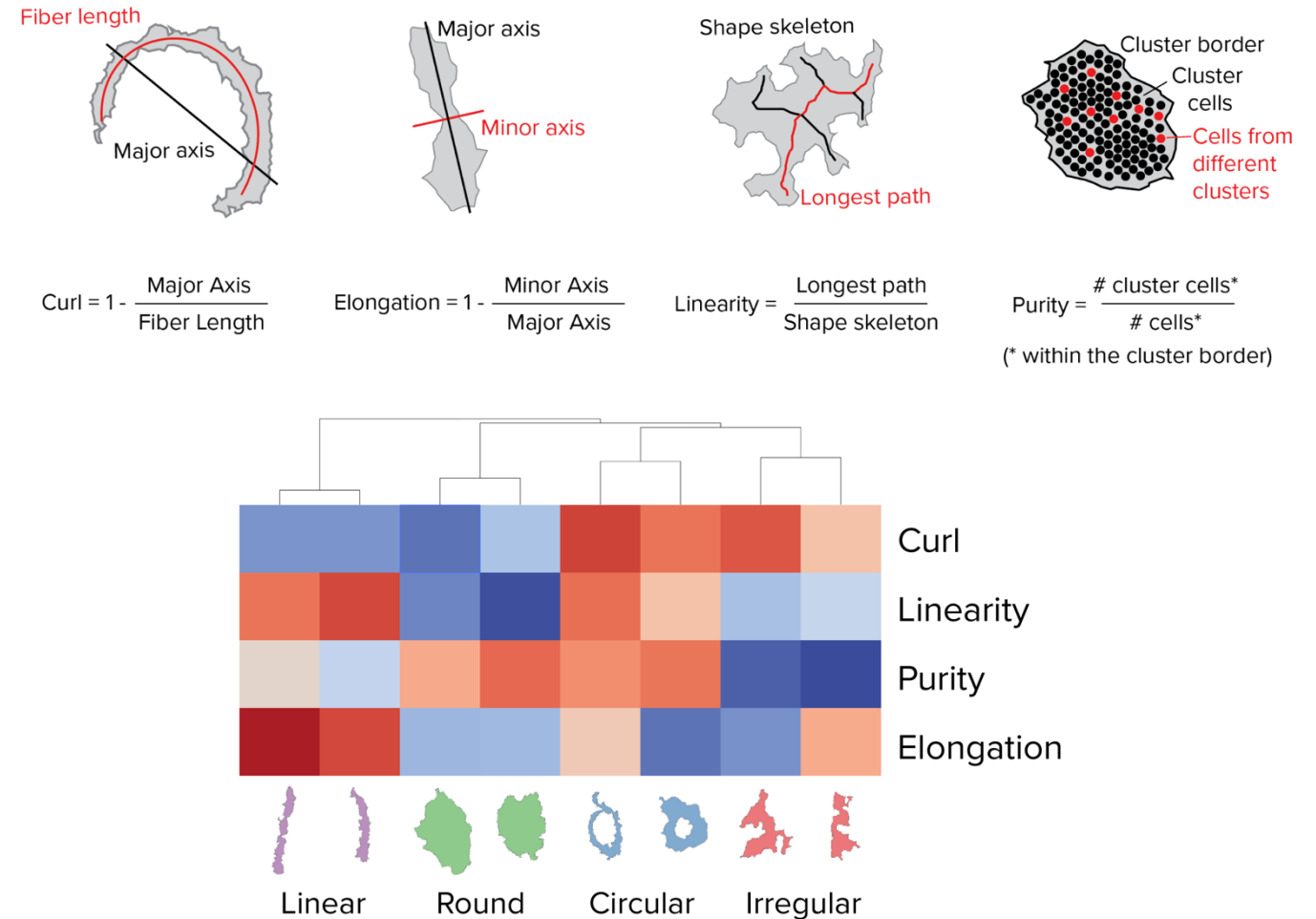
CellCharter



Transcriptomics-agnostic Spatial Domain (preliminary) analysis

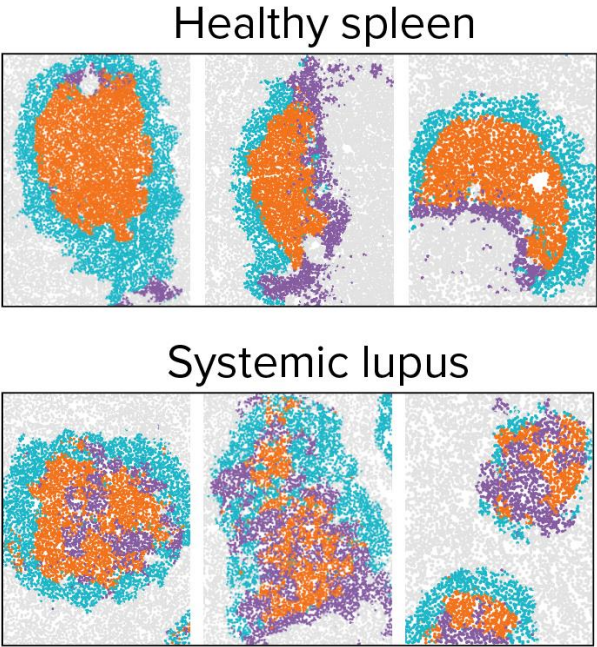
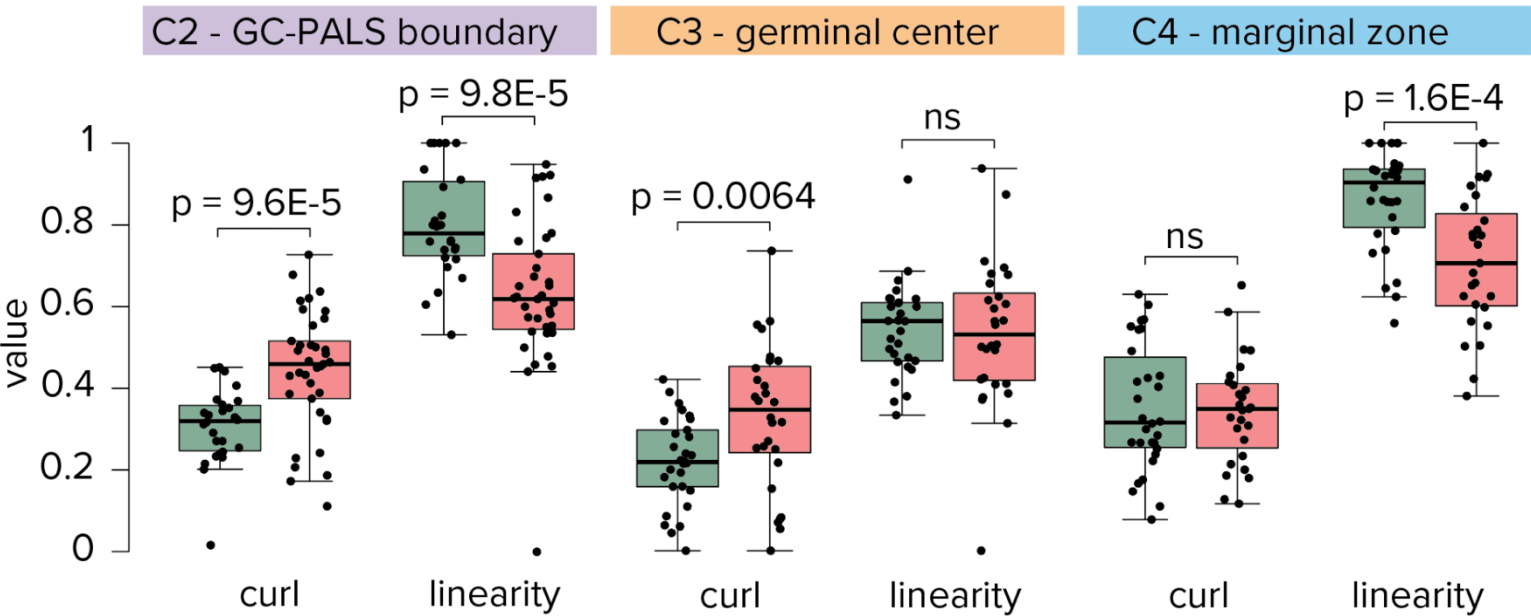
Shape metrics

- Curl: how curved or twisted a shape is
- Elongation: ratio of major and minor axis
- Linearity: how well a shape can be approximated by a linear path
- Purity: quantifies the fraction of cells within a cluster boundary that belong to that cluster



Transcriptomics-agnostic Spatial Domain (preliminary) analysis

Shape metrics



Which method is the best? → benchmark

JOURNAL ARTICLE

Benchmarking cell-type clustering methods for spatially resolved transcriptomics data

Andrew Cheng, Guanyu Hu | Analysis | Published: 15 March 2024

Briefings in Bioinformatics

<https://doi.org/10.1093/bi>

Published: 21 November

Benchmarking spatial clustering methods with spatially resolved transcriptomics data

Zhiyuan Yuan , Fangyuan Zhao, Senlin Lin

Nature Methods **21**, 712–722 (2024) | [Cite this article](#)

13k Accesses | **5** Citations | **45** Altmetric

Research | [Open access](#) | Published: 09 August 2024

Benchmarking clustering, alignment, and integration methods for spatial transcriptomics

Yunfei Hu, Manfei Xie, Yikang Li, Mingxing Rao, Wenjun Shen, Can Luo, Haoran Qin, Jihoon Baek & Xin Maizie Zhou 

Genome Biology **25**, Article number: 212 (2024) | [Cite this article](#)

3804 Accesses | **2** Citations | **20** Altmetric | [Metrics](#)

Can we use these existing benchmarks in some way?

Spatially aware clustering methods

LIBD DLPFC

Human brain

12 samples



- Used by almost all methods
- Method papers compare to older methods

Method X
Method X on LIBD

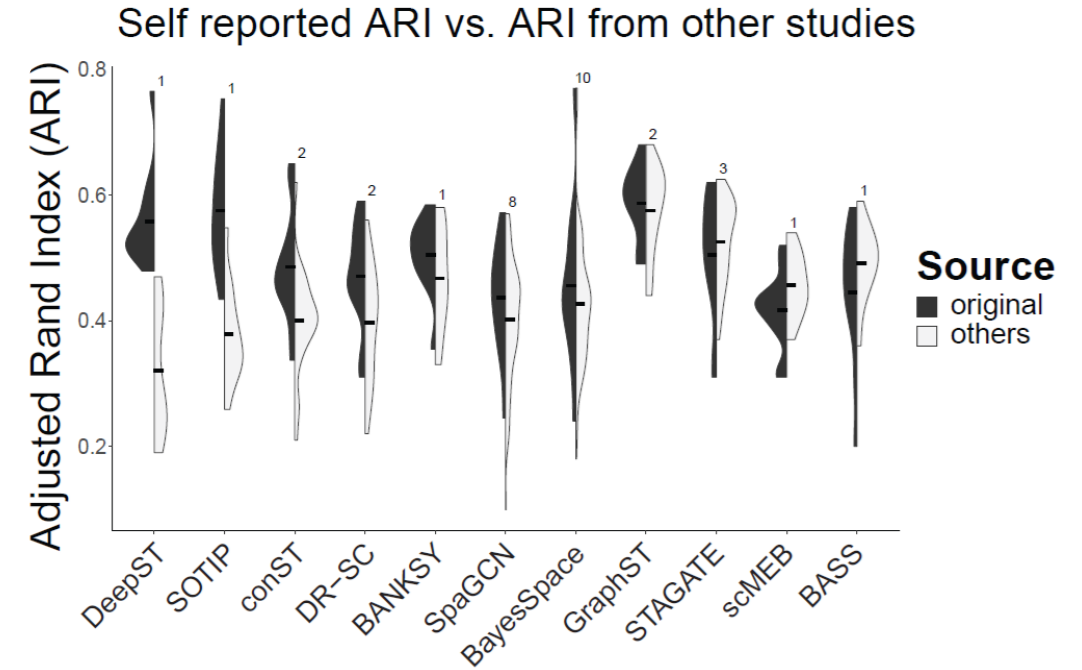
Method Y
Method Y on LIBD
Method X on LIBD

Method Z
Method Z on LIBD
Method Y on LIBD
Method X on LIBD

Benchmark
Method X on LIBD
Method Y on LIBD
Method Z on LIBD

These results should be identical... right?

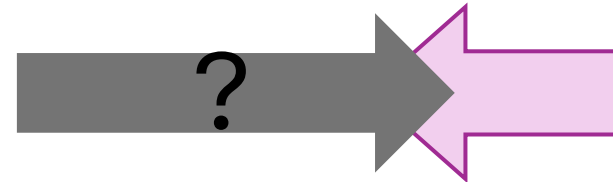
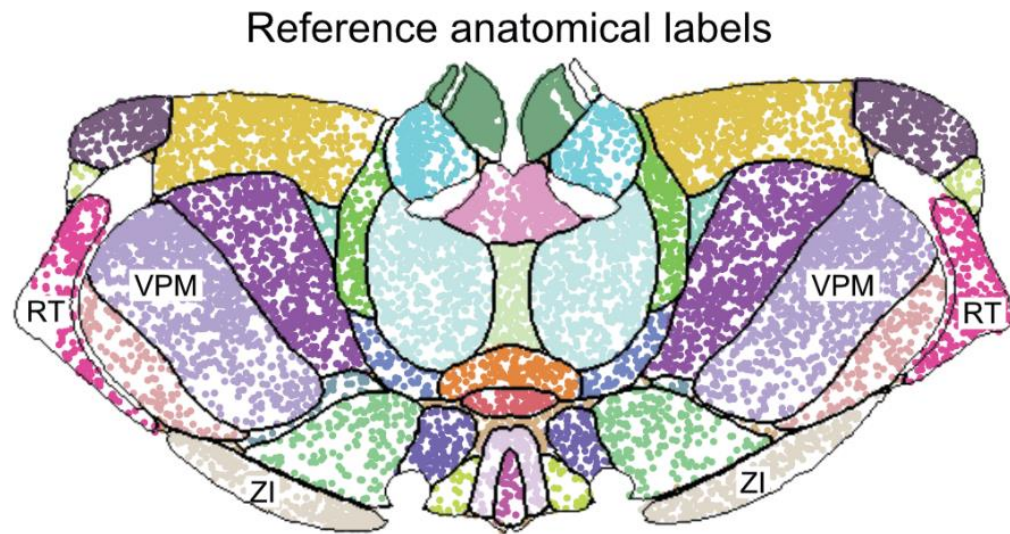
Use existing benchmarks



1. What causes this variability? → hyperparameters, preprocessing, ...
2. Are we evaluating the methods correctly?

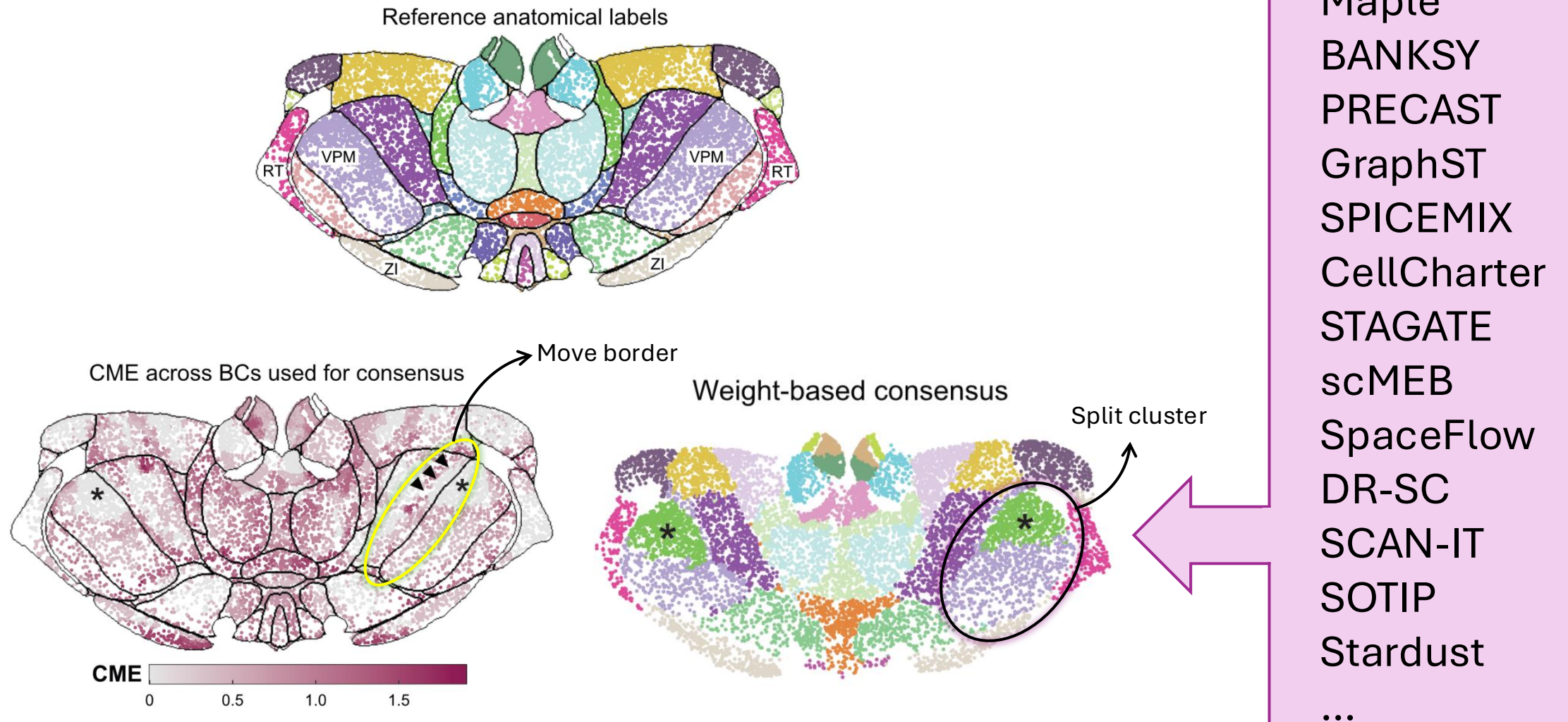
Reliance on the ground truth

- Is the ground truth meant to rank these methods?



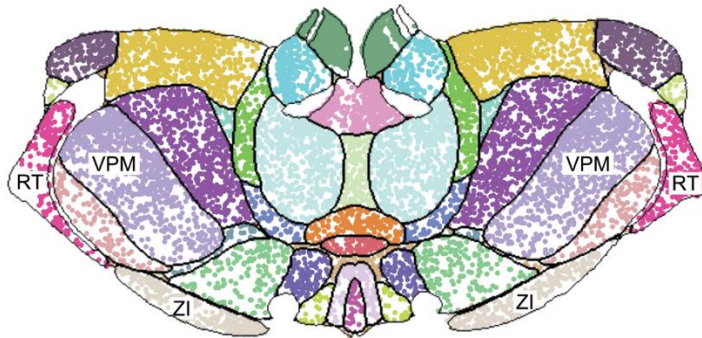
Maple
BANKSY
PRECAST
GraphST
SPICEMIX
CellCharter
STAGATE
scMEB
SpaceFlow
DR-SC
SCAN-IT
SOTIP
Stardust
...

Clustering as an **iterative** process between **methods** and **experts** with a **consensus**

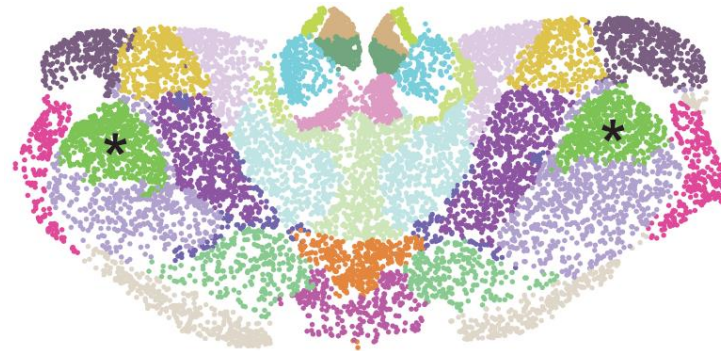


Redefining anatomical boundaries using consensus spatial clustering

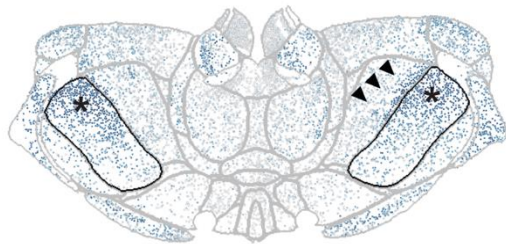
Allen Atlas



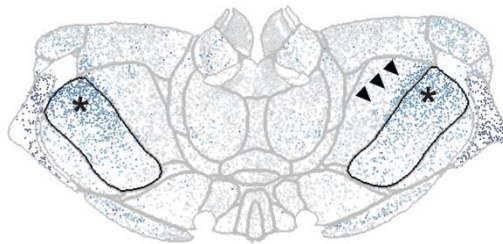
Consensus clustering



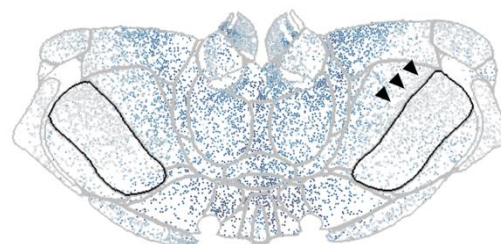
Kcnab3



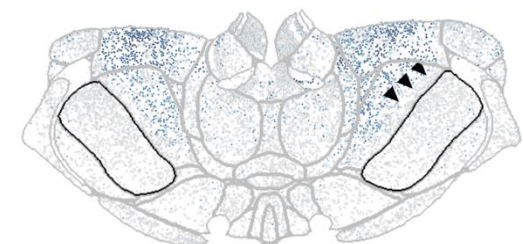
Pvalb



Calb1

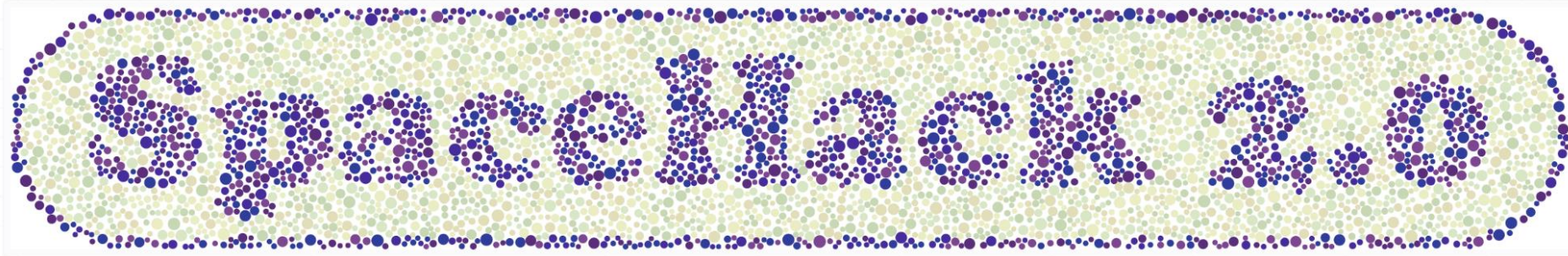


Tnc



Gene expression (log2cpt) 0 5

SACCELERATOR



SACCELERATOR

SACCELERATOR is the outcome of SpaceHack 2.0, a community-driven project to (not) benchmark domain identification methods for spatially-resolved transcriptomics data.

<https://spatialhackathon.github.io/SACCELERATOR/>

Summary

- Many definitions of domains, niches, spatial clusters.... -> **Spatially-aware clustering (SAC)**
- SAC usually relies on graphs to encode spatial distances and transcriptional/morphological similarity between neighboring cells
- Benchmarking is tricky! Think about the aim of clustering and how it fits your research question. There is no right answer to clustering (in general an ill-defined problem)

Thank You!

a.mahfouz@lumc.nl

www.mahfouzlab.org